

Applying Wav2vec2 to Improve Miscue Detection in Oral Reading Fluency (ORF) Assessment

Submitted in partial fulfilment of the requirements
for the award of the degree of

MS by Research

by

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June, 2025

Dedicated to my Parents & Family

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The results contained in this thesis have not been submitted in part or in full to any other University or Institute for the award of any degree or diploma to the best of our knowledge.

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Abstract

Automatic Speech Recognition (ASR) systems have emerged as a critical tool in educational contexts, particularly for assessing Oral Reading Fluency (ORF). However, traditional ASR approaches often underreport miscues due to strong language model constraints, thereby inflating reading accuracy scores and providing misleading feedback. This work investigates the application of the Wav2vec2 model, a self-supervised end-to-end speech recognition framework, to enhance miscue detection accuracy in ORF assessments.

We propose a two-pass system that combines the advantages of hybrid ASR models, known for easily customisable language modelling and better audio-to-text alignments, with Wav2vec2’s superior acoustic sensitivity. In the first pass, a hybrid ASR aligns the student’s oral reading with canonical text, identifying potential miscues using phonetically oriented alignment. The second pass uses Wav2vec2 to generate phoneme-level predictions and confidence scores, enabling refined detection of miscues initially missed due to language modeling biases.

Leveraging pseudo-labeling and cross-lingual training with multilingual datasets effectively adapts the Wav2vec2 model to children’s speech patterns. Specifically, we utilize unlabelled children’s speech data through pseudo-labeling and limited labelled children’s data, addressing the scarcity of annotated children’s speech and enhancing model adaptation to the target domain. Cross-lingual training is conducted using a phonetic superset that encompasses all possible phonemes across the involved languages, enabling robust multilingual acoustic modeling. This approach led to improved PER performance over baseline systems. In the second stage, word-level confidence features are calculated on aligned audio segments of words deemed correctly uttered by the first-pass ASR, and classification is applied to identify miscues, crucial for accurately capturing pronunciation variability and errors. Experiments on Hindi and English children’s speech datasets show substantial improvements in miscue detection. The first-pass miss rates were 72% (MPS English), 71% (KV-2000 Hindi), and 55% (KV-812 Hindi), which were reduced to 31%, 42%, and 33% respectively in the second pass through fusion of multiple Wav2vec2 and hybrid ASR features at a fixed 5% false positive rate (FPR).

The results underscore the effectiveness of combining acoustic modeling with confidence-based scoring to significantly reduce miscue miss rates while maintaining low false-positive rates. This research thus demonstrates the potential of Wav2vec2 to advance automated reading assessments, enabling more precise, reliable, and pedagogically valuable feedback in multilingual educational settings.

Contents

Approval	ii
Certificate	iii
Declaration	iv
Acknowledgements	v
Abstract	vi
List of Figures	ix
List of Tables	xi
1 Introduction	1
2 Literature Survey	5
2.1 Introduction to Wav2vec2	6
2.1.1 Architecture	6
2.1.2 Wav2vec2 Variants	8
2.2 Multilingual Speech Recognition	11
2.2.1 Introduction	11
2.2.2 Multilingual Training Strategies and Loss Functions	13
2.2.3 Benchmark : FLEURS Dataset	16
2.3 ASR Architectures for Reading Miscue Detection	17
2.3.1 Hybrid ASR Systems	18
2.3.2 End-to-End ASR with Wav2vec2	18
2.3.3 Whisper and its Limitations	19
2.3.4 Hybrid + Wav2vec2: Leveraging the Best of Both	20
2.4 Relevant Literature	21
2.4.1 Phone vs. Grapheme Recognition in E2E ASR	21
2.4.2 Wav2vec2 for Mispronunciation Detection	23
2.4.3 Children’s Speech Adaptation and Pseudo-labeling	25
2.4.4 Word-Level Confidence Estimation in End-to-End ASR	26

3	Datasets and Characteristics	29
3.1	Sources	29
3.2	Training Dataset	31
3.3	Test Datasets	35
4	A two-pass miscue detection system	43
4.1	First Pass: Hybrid ASR	44
4.2	First pass : Post-Processing Alignment	47
4.3	Second Pass: Confidence Feature Calculation	49
4.4	Feature Fusion and Classification	51
5	Building Wav2vec2 Phone Recognition model	53
5.1	Acoustic Model (AM) Training	53
5.2	Inference and Phone Sequence Prediction	57
5.3	Pseudo Label Generation (Unlabeled Data)	58
5.4	Language Model Integration	61
6	English language - Wav2vec2 Model	67
6.1	Model Training Setup	67
6.2	Results and Analysis	70
7	Hindi language - Wav2vec2 Multilingual Model	73
7.1	Multilingual Fine-Tuning Experiments	74
7.2	Pseudo Label Generation, Filtering and Experiments	76
7.3	Experiment of Adding Hindi Labelled Data of Children	84
7.4	Comparison of Wav2Vec2 and Hybrid ASR	88
8	Confidence Feature	91
8.1	Levenshtein Distance: LevDist and CostLevDist	91
8.2	End-to-End Confidence Features	93
8.3	Experiments	94
9	Miscue Detection Performance	99
10	Discussion and Analysis	109
11	Conclusion and Future Work	119
11.1	Conclusion	119
11.2	Future Work	121

Publications and Resources

References

List of Figures

1.1	Automatic reading assessment system	2
1.2	Three types of miscues	2
1.3	Missed mispronunciations by ASR due to limited vocabulary and grammar constraints	3
2.1	Wav2Vec2 Architecture	7
3.1	Distribution of % Miscue Rate for MPS dataset - English	37
3.2	Distribution of % Miscue Rate for KV-2000	39
3.3	Distribution of % Miscue Rate for KV-812	41
4.1	Overview of the proposed 2-pass miscue detection system.	44
5.1	Wav2vec2-Large Phone Recognition model architecture	54
5.2	WER (%) heatmap from initial grid search over a broad range of hyperparameters: $\alpha \in [-4, 4]$, $\beta \in [0, 5]$. Significant variation observed with small parameter changes.	63
5.3	WER (%) heatmap from refined search: $\alpha \in [-1.5, 1.5]$, $\beta \in [0, 7.5]$ (step = 0.5). Best results clustered around $\alpha = 0.5$	63
5.4	Fine-grained heatmap showing best WER of 20.56% at $\alpha = 0.4$, $\beta = 5.0$. Used for final beam size experiment.	64
7.1	Distribution of CTC Loss values across the unlabelled Hindi children's dataset. Most utterances lie around 0.1.	78
7.2	Distribution of PRC scores from beam search decoding across the unlabelled Hindi children's dataset. Most scores are concentrated near zero.	78
7.3	CTC Loss vs. PER. A strong positive correlation (0.84) indicates CTC loss is a reliable predictor of transcription quality.	79
7.4	PRC Score vs. PER. A weaker negative correlation (-0.08) is observed.	80
8.1	Phone sequence extraction and Levenshtein distance computation using Wav2vec2 outputs and hybrid ASR alignment.	92
8.2	End-to-end word-level confidence feature extraction pipeline from Wav2vec2 model outputs.	93

9.1	Detection Error Tradeoff (DET) curve illustrating miscue detection performance on the MPS English dataset. Each curve represents a different confidence feature or feature fusion method evaluated in the second-pass system. The curves are plotted with Miss Rate (False Negative Rate) against False Positive Rate (FPR), with a focus on performance at the operational point of 5% FPR.	102
9.2	Detection Error Tradeoff (DET) curve comparing different features for miscue detection on the KV-2000 Hindi test set. Fusion-All achieves the lowest miss rate at 5% FPR, indicating the benefit of combining features from both Hybrid ASR and Wav2vec2-based systems.	105
9.3	DET curve showing miscue detection performance for the KV-812 Hindi test set. Similar to KV-2000, Fusion-All outperforms individual features and fusion-best configurations, demonstrating robust generalization across different testset.	106
10.1	Confusion matrix of phone predictions on the MPS English dataset. .	114
10.2	Confusion matrix of phone predictions on the KV-2000 Hindi dataset. .	115
10.3	1st : Word boundary, 2nd : Phone boundary from Hybrid ASR system and 3rd : Wav2vec2 Prediction boundary	117

List of Tables

2.1	Comparison of ASR Architectures for Reading Miscue Detection. . . .	21
3.1	Speech Datasets Sources and its characteristics	29
3.2	Summary of English language datasets used for training and evaluation.	34
3.3	Summary of Hindi-Marathi datasets with utterance, hour, and speaker counts used for training and testing.	35
3.4	Distribution of word reading errors across the 1,10,898 attempted words in MPS dataset	37
3.5	Distribution of word reading errors across the 1,28,955 attempted words in KV-2000 dataset	39
3.6	Distribution of word reading errors across the 64427 attempted words in KV-812 dataset	40
4.1	WER and miscue detection performance of Hybrid ASR with different LM constraints (MPS English)	45
4.2	WER and miscue detection performance of Hybrid ASR with different LM constraints (KV 2000 - Hindi)	46
4.3	WER and miscue detection performance of Hybrid ASR with different LM constraints (KV 812 - Hindi)	46
5.1	WER (%) Comparison Across Model Configurations for Two Hindi Datasets	65
6.1	Phone Error Rate (PER) on the MPS English test set for different ASR systems and training strategies. The "Combined" setting refers to training on both IITM and WPP datasets from scratch using the SSL model. The "Fine-tuning" setting indicates continued training of a model first trained on IITM data and then fine-tuned on WPP data. TL: Transformer Layer, Conv: Feature Convolution layer	70

7.1	PER (%) on KV-2000 Hindi using XLSR-based Wav2vec2 model fine-tuned with various strategies. Combining IITM and AIS datasets yields better performance than sequential fine-tuning. Where arrow $\rightarrow$ shows further fine-tuning and + shows combining the dataset. TL: Transformer	75
7.2	PER (%) on KV -2000 Hindi comparing XLSR and AI4Bharat IndicWav2vec2 models under various training configurations. Fine-tuning the convolutional layer (Conv) along with Transformer layers significantly improves performance on children’s speech.	76
7.3	PER (%) on KV 2000 - Hindi comparing baseline models with those trained using pseudo-labeled data (Baseline and Midline - 2025) filtered by CTC and PRC. PRC-based filtering yields the best performance.	83
7.4	Ablation results using PRC-filtered pseudo-labeled data. B: Baseline (12.05 hrs), M: Midline (11.82 hrs), E: Endline (12.12 hrs) of 2024-25 assessment pseudo labelled data. Adding more diverse and high-quality children’s speech improves PER.	84
7.5	PER (%) comparison of different models on KV2000 (overlapping) and KV-812 (non-overlapping) Hindi test sets. B: Baseline (12.05 hrs), M: Midline (11.82 hrs), E: Endline (12.12 hrs) of 2024-25 assessment pseudo labelled data and KV Train Hindi dataset of 23 hours.	86
7.6	PER (%) comparison of Hybrid ASR and Wav2Vec2 models on KV 2000 (overlapping) and KV 812 (non-overlapping) Hindi test sets. B: Baseline (12.05 hrs), M: Midline (11.82 hrs), E: Endline (12.12 hrs) pseudo-labeled data. KV Labeled Hindi dataset: 23 hours.	88
8.1	Comparison of confidence-based feature performance using Miss Rate (at 5% FPR) and AUC on the English MPS test dataset.	95
8.2	Comparison of confidence-based feature performance using Miss Rate (at 5% FPR) and AUC across Hindi KV-2000 and KV 812 test datasets.	96
9.1	English MPS Test Dataset : Miss rate at FPR=5% and AUC for features across different classes and their combinations. Bold font: best in class, (R: raw; T: temperature, L: logmax, E: entropy; S: sum)	101
9.2	Hindi KV-2000 and KV-812 Test Datasets: Miss rate and AUC at FPR=5% for different features and their combinations on Prompt LM. Bold font: best in class (within each dataset).	104

10.1	PER (%) by Prediction Type after 2-stage Reading miscue System, where only correctly marked words from the first-pass system are forwarded to the second pass. Overall PER on KV-2000 Hindi and MPS English is 4.45 and 9.67 respectively. Hindi model is AI4B SSL IITM + AIS + Pseudolabelled_PRC (B+M+E) → KV Train Hindi (Fine-tuned) as in table 7.6 and English Model is Wav2vec2-XLSR (IITM → WPP (only TL) fine-tuning) as in table 6.1. Cell values show PER and the proportion of that prediction type in the test set (%)	116
11.1	Phone mapping from the IITB superset to the IITM English and IITM Hindi phone sets.	

Chapter 1

Introduction

A child's reading ability is an important part of learning for various reasons, as it serves as a foundational skill that influences many aspects of their academic, personal and future professional life. With proper training and feedback this ability can be improved. Assessment of children's reading ability using an Automatic Speech Recognition (ASR) system can offer several advantages in an educational context [White et al. \(2021\)](#). It is more efficient and is particularly beneficial in schools with low accessibility to resources, allowing educators to allocate more time to teaching and intervention strategies. This also provides a standardized and consistent method of evaluating reading ability across different students, classrooms, and schools. This standardization contributes to more reliable and comparable results. The evaluation metric is Words Correct Per Minute (WCPM), which measures the number of correctly read words by a reader within one minute.

As shown in Figure [1.1](#) to generate WCPM score the ASR decoded text is aligned with the canonical text (prompted text for reading) to get a report of total miscues. Miscues can be of three types:

- **Insertion:** Where a word absent in canonical text is spoken by the reader.

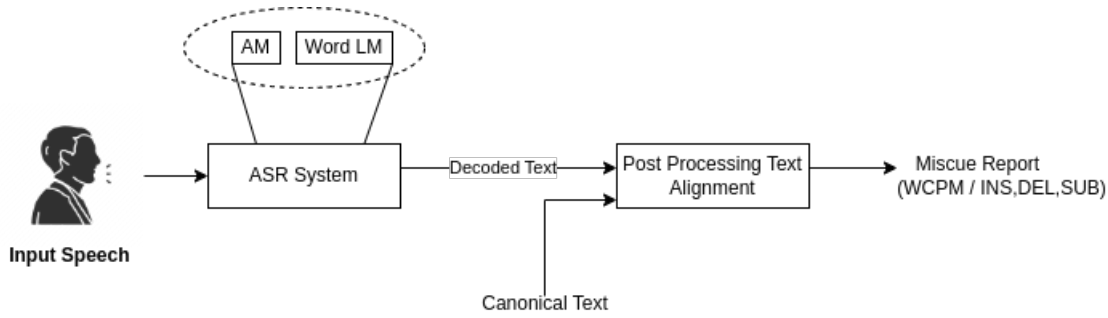


FIGURE 1.1: Automatic reading assessment system

- **Deletion:** When a word in canonical text is omitted by the reader.
- **Substitution:** When a word in canonical text is replaced by another word.

Figure 1.2 shows an example of the above mentioned miscues.

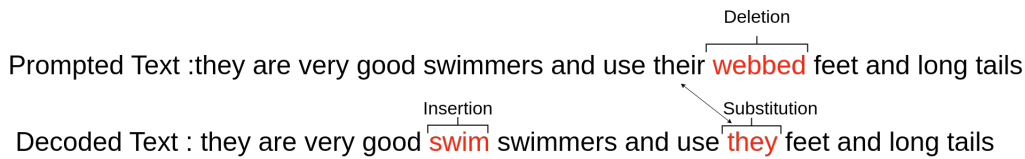


FIGURE 1.2: Three types of miscues

In field it has been observed that the WCPM reported by an ASR System is highly inflated. A human listener will report more miscues than detected by an ASR System [Mostow \(2012\)](#).

In our system, a hybrid ASR model is paired with a constrained language model (LM) trained on the prompt (story) text. This restricted vocabulary helps achieve lower Word Error Rate (WER) and significantly reduces false alarms. Using a general LM trained on large corpora introduces a broader vocabulary, increasing the risk of decoding words not present in the prompt but acoustically similar—especially problematic with children’s speech, which is often inconsistent. This can lead to correctly pronounced words being falsely flagged as miscues, which negatively

impacts the perceived performance of students and the credibility of the system in educational settings. Therefore, using a limited vocabulary LM aligned with the expected prompt is generally more reliable for accurate miscue detection.

Canonical Text : Food **is** scarce. If they have nothing to eat for **several** days ...

Manual Transcription : food **eats** scarce if they have nothing to eat for **सेरवल** days

Decoded Text : food **is** scarce if they have nothing to eat for **several** days

FIGURE 1.3: Missed mispronunciations by ASR due to limited vocabulary and grammar constraints

In Figure 1.3 the word "is" is more likely to be followed by "food" therefore "is" gets decoded in place of "eats". A Devanagari word is used by transcribers where an illegible word is present. In such cases, obviously the word is not present in the vocabulary and some other close words get decoded. Due to these limitations, it is important to verify whether a decoded word aligned with the canonical word and labeled as "correct" was actually pronounced correctly. Moreover, beyond identifying whether a canonical word was mispronounced, it is also valuable to obtain an accurate phone-level transcription of what was actually uttered. Since children's reading errors often involve substitutions, insertions, or deletions of small sound units (phones or syllables), accurate phone recognition is essential for capturing such fine-grained miscues.

To deepen our understanding of children's reading patterns, this study could be helpful to investigate the differences in the types of errors made by children while reading Hindi and English. While error categorization research has been extensively conducted for English, similar studies for Hindi remain largely unexplored. Previous research, such as the work [Savage et al. \(2001\)](#), identified four primary categories of errors in English: errors with orthographic overlap, errors preserving the initial phoneme, errors preserving the final phoneme, and scaffolding errors, which preserve

both the initial and final phonemes. Furthermore, Ehri’s categories for English reading errors provide additional frameworks for analyzing miscue types [Ehri \(2005\)](#).

Examples of errors that can be investigated in this context include rime neighbors (e.g., rain/vain), substitution neighbors (e.g., rain/ruin), and transposition neighbors (e.g., clam/calm). Additionally, scaffolding errors that reflect phonetic similarities, such as target/tight in English, and word/non-word confusions offer valuable insights into error patterns [Stuart and Coltheart \(1988\)](#); [Savage et al. \(2001\)](#).

This comparative analysis between Hindi and English reading errors can enrich error classification frameworks, especially for underrepresented languages like Hindi, while contributing to improved systems for reading miscue detection. Such a study motivates the need for a system with low PER for any language used in the ORF assessment.

Chapter 2

Literature Survey

In this chapter, we provide a comprehensive survey of the literature related to automatic speech recognition (ASR) technologies keeping in mind our task of reading miscue detection. Motivated by the need for accurate phone recognition with limited training data, We begin by introducing the Wav2vec2 model and its architectural variants, exploring Multilingual speech recognition’s architecture, training strategies and loss functions. This is followed by an overview of ASR systems that have been applied or adapted for reading miscue detection task—including traditional hybrid models, end-to-end approaches based on Wav2vec2, and recent models like Whisper. We also examine approaches that combine hybrid and Wav2vec2 architectures to leverage their complementary strengths in miscue detection. The chapter concludes with a review of key research themes such as phoneme- vs. grapheme-based recognition, mispronunciation detection using Wav2vec2, adaptation to children’s speech using pseudo-labeling, and word-level confidence estimation in end-to-end ASR.

2.1 Introduction to Wav2vec2

Recent advances in self-supervised learning (SSL) have significantly improved performance in automatic speech recognition (ASR), especially in low-resource and multilingual settings. One of the most influential models in this domain is Wav2vec 2.0 (Wav2vec2), introduced by [Baevski et al. \(2020\)](#). This model learns robust speech representations directly from raw audio without relying on transcribed data during the pretraining phase, making it highly suitable for downstream applications and under-resourced languages.

2.1.1 Architecture

Traditional ASR systems require large quantities of labeled (transcribed) speech data, which is expensive and time-consuming to obtain, particularly for low-resource languages. Wav2vec2 mitigates this issue by leveraging large amounts of unlabeled audio to learn general-purpose speech representations by self-supervised learning (SSL) pre-training, which can later be fine-tuned with a relatively small amount of labeled data.

The Wav2vec2 model consists of three primary components as in figure [2.1](#):

- **Feature Encoder (CNN)** : This block takes raw audio waveform as input and converts it into a sequence of latent speech representations. It consists of several convolutional layers that reduce the input dimensionality and capture low-level acoustic patterns.

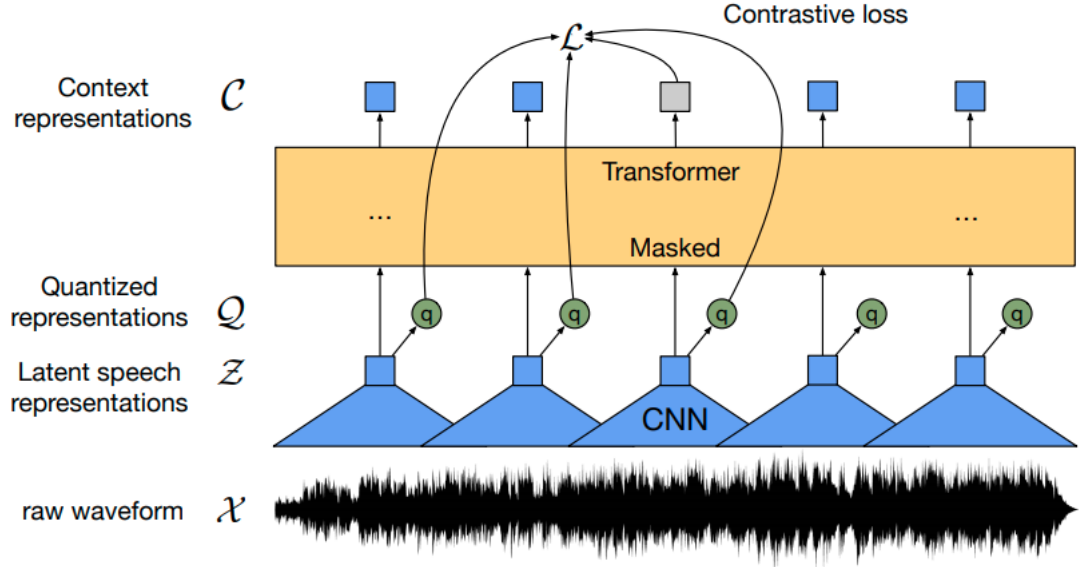


FIGURE 2.1: Wav2Vec2 Architecture

- **Contextual Transformer Network** : The latent representations are fed into a Transformer encoder [Vaswani et al. \(2017\)](#) that models long-range dependencies in the audio. This block outputs context-rich embeddings that are sensitive to phonetic and prosodic information.
- **Quantization Module (during pretraining only)** : In the self-supervised pretraining phase, a quantization module discretizes the encoder output into a finite set of codebook entries (akin to subword units), forming the basis for the contrastive learning task.
- **Pretraining Objective** : Wav2vec2 employs a **contrastive loss** to train the model in a self-supervised manner. The training process involves:
 - Masking spans of the latent feature sequence.
 - Predicting the correct quantized vector for the masked region from a set of distractors (negative samples).

- Optimizing the model to distinguish the correct future token from incorrect ones, forcing the model to learn meaningful representations.

This approach is inspired by BERT-style pretraining [Devlin et al. \(2019\)](#) in NLP but adapted for continuous audio streams. Once pretrained, the model can be fine-tuned for various supervised downstream tasks such as:

- Automatic Speech Recognition
- Phone Recognition
- Keyword Spotting
- Speech Translation

For ASR, a Connectionist Temporal Classification (CTC) layer [Graves et al. \(2006\)](#) is typically added on top of the Transformer encoder. The model is then fine-tuned using labeled audio-transcription pairs. Further technical details on training for specific tasks, such as phone recognition and confidence modeling, will be discussed in subsequent chapters.

2.1.2 Wav2vec2 Variants

The Wav2vec2 architecture has inspired several variants, each tailored to specific datasets, languages (English, Multilingual or Indic), and SSL training objectives during self-supervised pretraining. Below is an overview of notable SSL Wav2vec2-based models for English and Indian languages:

- **Wav2vec2-Base and Wav2vec2-Large:** Wav2vec 2.0 masks the speech input in the latent space and solves a contrastive task defined over a quantization of the latent representations, which are jointly learned during training [Baevski et al. \(2020\)](#). The model was pretrained on approximately 53k hours of unlabeled speech data and later fine-tuned on 960 hours of labeled English speech from the LibriSpeech dataset. There are two main versions of this model: Wav2vec2-Base, containing 95 million parameters, and Wav2vec2-Large, with 317 million parameters.

Both serve as foundational models for English ASR tasks, with the large variant offering better performance due to its higher capacity.

- **XLSR (Cross-Lingual Speech Representations):** XLSR-53 extends the Wav2vec2.0 framework to the multilingual setting by jointly pretraining on speech data from multiple languages. The model is trained using a contrastive self-supervised learning objective over masked latent speech representations, similar to Wav2vec2. A shared quantization module learns discrete speech units across languages, allowing the Transformer to learn language-independent speech representations [Conneau et al. \(2021\)](#). This enables knowledge transfer between related and unrelated languages, which is especially beneficial for low-resource languages.

The XLSR-53 model is pretrained on 56,000 hours of speech data from 53 languages, including few Indian languages (Tamil, Assamese, Bengali). It uses datasets like MLS [Pratap et al. \(2020\)](#), CommonVoice [Ardila et al. \(2020\)](#), and BABEL [Gales et al. \(2014\)](#), providing broad phonetic and linguistic coverage. Based on the Wav2vec2-Large configuration, XLSR-53 includes 317 million parameters. It uses the same convolutional feature encoder and Transformer

backbone, along with a shared multilingual quantizer to align speech units across languages.

- **HuBERT (Hidden-Unit BERT):** HuBERT introduces a two-stage pre-training approach. It uses offline k-means clustering to assign discrete labels (or "hidden units") to audio features. These labels serve as targets for a BERT-style masked prediction loss, applied only over masked regions. This forces the model to learn both acoustic and contextual (language-like) structure from raw speech [Hsu et al. \(2021\)](#). HuBERT is trained exclusively on English data. The Base model uses 960 hours of LibriSpeech, while the Large and Extra-Large variants are pretrained on 60,000 hours of Libri-Light. HuBERT uses the same core architecture as Wav2vec2: Base (95M parameters), Large (316M parameters) and Extra-Large (1B parameters). The key difference lies in the target generation and loss function, which relies on clustered units rather than quantized codebooks.
- **IndicWav2vec2:** IndicWav2Vec follows the XLSR-Wav2vec2 self-supervised learning framework, pretraining on raw audio to learn robust acoustic representations for Indian languages [Javed et al. \(2022\)](#). The model is pretrained on 17,000 hours of raw speech from 40 Indian languages, including English, across diverse domains like education, news, and finance. It is fine-tuned for downstream ASR on 9 Indian languages using datasets such as MSR [Srivastava et al. \(2018\)](#), MUCS [Diwan et al. \(2021\)](#), and OpenSLR [Kjartansson et al. \(2018\)](#).

2.2 Multilingual Speech Recognition

Multilingual speech recognition has advanced rapidly in recent years, expanding ASR beyond high-resource languages. Traditional ASR systems required separate models and lexicons per language, but end-to-end (E2E) approaches now enable a single model to transcribe multiple languages.

2.2.1 Introduction

Modern multilingual ASR builds on several core E2E architectures have emerged in the last decade. All these approaches learn to map audio sequences to text sequences directly, but they differ in how they model sequence alignment and decoding. The **encoder** is usually a Transformer or, more recently, a *Conformer* that interleaves self-attention and depth-wise convolutions to capture both global context and local features [Gulati et al. \(2020\)](#). Built on this encoder are three canonical decoding approaches:

1. **Seq2Seq with Attention (LAS)**. One popular E2E approach is the sequence-to-sequence encoder-decoder model with attention (often called Listen, Attend, and Spell). An encoder (e.g. bidirectional LSTM or Transformer) processes the input audio features, and a decoder (often an RNN or Transformer) generates the transcript one token at a time, attending to the encoder’s outputs. This architecture was first applied to ASR around 2015 and proved that a single neural network could learn to align and transcribe speech without an explicit pronunciation lexicon [Chan et al. \(2015\)](#). Attention-based models can handle long context and naturally model multilingual output by using a

shared token vocabulary across languages. However, they are typically non-streaming (requiring the full utterance) and the decoder’s built-in language modeling means they may need large text exposure to cover many languages and sensitive to vocabulary size.

2. **Connectionist Temporal Classification (CTC).** CTC introduced by [Graves et al. \(2006\)](#) provides an alignment-free training criterion for ASR. An encoder (often a deep CNN, LSTM or Transformer) produces a sequence of frame-wise scores over characters or phonemes, and CTC computes the probability of all alignments that spell out a given transcript (using a special blank symbol for “no output”). The model is trained to maximize the total probability of the correct transcription, implicitly learning an alignment. CTC-based models directly output characters or word pieces, making them attractive for multilingual ASR because we can use a universal character set (e.g. Unicode characters or bytes covering all languages). There’s no separate decoder network or language model built-in – instead, an external language model can be applied during beam search decoding to improve WER. This simplicity makes CTC a common choice for multilingual systems. For example, Meta’s MMS and XLSR models use a transformer encoder with a CTC loss over characters [Conneau et al. \(2021\)](#); [Pratap et al. \(2024\)](#). One downside is that CTC assumes conditionally independent token outputs given the encoder, which may limit accuracy on long contexts, but it is often mitigated by coupling with an external LM.
3. **RNN-Transducer (RNN-T).** The RNN-T architecture [Graves \(2012\)](#) was widely adopted in industry for streaming ASR. An RNN-T consists of an encoder network, a prediction network (a neural LM that processes the previously

output tokens), and a joint network that combines encoder and prediction features to output the next token. Training uses a transducer loss that sums over all valid alignments, similar to CTC but with the prediction network assisting. RNN-T effectively integrates an internal language model (the prediction network) into the ASR model. This is beneficial for multilingual ASR because the prediction network can learn cross-lingual language patterns if trained on mixed-language text. Google’s large-scale multilingual models (e.g. USM) use an RNN-T with a Conformer encoder [Zhang et al. \(2023\)](#). RNN-T can operate in streaming mode (outputting words as audio comes in) and tends to handle long audio well. However, it’s more complex to train than CTC or attention models, and the internal LM can cause domain or language biases if not carefully managed (for example, favoring high-resource language vocabulary).

2.2.2 Multilingual Training Strategies and Loss Functions

Scaling ASR to many languages introduces challenges beyond the core model architecture. Recent research has developed strategies for training data, initialization, and loss functions to improve multilingual performance:

- **Multilingual Self-Supervised Pretraining:** A major breakthrough enabling multilingual ASR was self-supervised learning on unlabeled audio from many languages. Facebook’s XLSR (Cross-Lingual Speech Representations) approach extended the wav2vec 2.0 model to dozens of languages [Conneau et al. \(2021\)](#). These pre-trained models are publicly available and have become a foundation for many multilingual ASR systems. The strategy is to pretrain a generic audio encoder on massive multilingual audio, using losses like contrastive InfoNCE (as in wav2vec 2.0) or masked prediction, and then

fine-tune on labeled data. This pretraining yields universal acoustic features that transfer across language. Self-supervised pretraining thus drastically reduced the amount of transcribed data needed for new languages and enabled one model to cover many languages.

- **Multilingual Fine-Tuning:** Given a pre-trained encoder, one can fine-tune it on multiple languages at once (multilingual fine-tuning) or on one language at a time. Jointly training on many languages' transcribed data tends to improve cross-lingual transfer – low-resource languages benefit from the shared representations learned from high-resource languages. However, naively mixing languages can lead to dominant languages crowding out weaker ones - Solutions include upsampling low-resource languages, curriculum learning (start with high-resource then gradually add low-resource), and using language embedding features to prevent interference. Another technique is to use language-specific modules within a shared model. MMS introduced **Language-Specific Adapters and Heads (LSAH)**: small adapter layers and output layers for each language on top of a shared encoder [Pratap et al. \(2024\)](#). This allows specialization per language without training completely separate models. They found LSAH improved average error rates. Overall, multilingual fine-tuning is now standard: models like Whisper [Radford et al. \(2022\)](#) and USM [Zhang et al. \(2023\)](#) are trained on combined transcribed data from dozens of languages simultaneously. Recently Large language models with over 1B parameters are expensive to fine-tune for small tasks due to high computational costs. Low-Rank Adaptation (LoRA) offers a parameter-efficient alternative by updating only a few low-rank matrices, achieving performance close to full fine-tuning. This technique is now also being explored for large speech models to enable efficient adaptation.

- **Cross-Lingual Transfer Few-Shot Learning:**

One advantage of multilingual training is the ability to transfer to new languages with little data, or even zero-shot. Models with a shared acoustic space and character set can sometimes transcribe a new language out-of-the-box if it's related to trained languages. For instance, OpenAI's Whisper model (trained on 99 languages) can recognize some similar languages it wasn't explicitly trained on (with reduced accuracy). More commonly, we adapt a multilingual model to a new target language with a small amount of supervised data – a few-shot fine-tune. Cross-lingual transfer is especially effective when the languages share phonetic or script similarity. A striking result was reported by Meta MMS: by pre-training on 1,400+ languages, their model improved low-resource languages like Amharic, Lao, and Malayalam significantly (lower CER), although a small degradation in a few high-resource languages was observed due to capacity trade-off [Pratap et al. \(2024\)](#). Techniques like adapters and language-specific heads are used to mitigate this trade-off, maintaining performance on high-resource languages while adding many new ones.

- **Loss Functions & Multitask Objectives** End-to-end models are trained with one or more sequence-level loss functions. CTC loss (used in wav2vec/XLSR and MMS) aligns well with frame-based encoders and is language-agnostic. Transducer loss (RNN-T) combines acoustic and linguistic modeling; it effectively trains an implicit LM jointly with the acoustic model. This can be advantageous for multilingual ASR if large text corpora are available – for example, Google's USM incorporated unlabeled text (unpaired with audio) in pre-training to boost the RNN-T predictor model. On the other hand, sequence-to-sequence models use a cross-entropy loss on the decoder's output tokens like [Radford et al. \(2022\)](#), often along with an attention mechanism.

Some research combines objectives: e.g. a model might be trained with a weighted sum of CTC loss and seq2seq loss, leveraging CTC for alignment help and seq2seq for better language modeling. In multilingual settings, there may also be **auxiliary losses**: language identification loss (predicting a language ID from the encoder) can help the model learn language-specific cues and reduce confusion between similar languages. Some studies had tried to apply cross-entropy loss at frame level by generating forced aligned ground truth, but this kind of approach are not popular in community. Overall, the choice of loss ties into the architecture: CTC is popular for large multilingual models due to its simplicity and ease of decoding with external LMs, while RNN-T/attention losses integrate language modeling internally, which can be powerful if adequately trained (e.g. with massive text data).

2.2.3 Benchmark : FLEURS Dataset

The Few-shot Learning Evaluation of Universal Representations of Speech (FLEURS) dataset is a multilingual speech corpus comprising 102 languages. It contains a total of approximately 1,400 hours of audio, divided into: 987 hours for training, 120 hours for development (validation), and 283 hours for testing. The speech data consists of read speech, and the text used for recording was sourced from Wikipedia. Importantly, the dataset includes both parallel text and parallel speech, which makes it suitable for tasks involving cross-lingual and multilingual speech recognition, translation, and alignment [Conneau et al. \(2023\)](#).

- **w2v-BERT (2022)** [Chung et al. \(2021\)](#): 12.3 % CER.
- **Maestro-U (2022)** [Chen et al. \(2022\)](#): 8.7 % CER via cross-modal speech–text training.

- **USM (2023)** [Zhang et al. \(2023\)](#): 6.5 % CER using 12 M h SSL + 90 k h supervised + massive text.
- **MMS + LSAH + LM (2024)** [Pratap et al. \(2024\)](#): 6.3 % CER across 1107 languages.

A Transformer (or Conformer) encoder paired with either CTC or RNN-T decoding, pretrained on vast multilingual audio and fine-tuned with balanced supervision, currently delivers state-of-the-art recognition ($\approx 6\%$ CER on the 102-language FLEURS benchmark). The choice between CTC, Seq2Seq or RNN-T ultimately hinges on deployment constraints (streaming vs. offline) and the availability of text for language modelling.

2.3 ASR Architectures for Reading Miscue Detection

Automatic Speech Recognition (ASR) plays a crucial role in evaluating oral reading fluency, particularly in detecting reading miscues such as insertions, deletions, and substitutions. In this context, the choice of ASR architecture has a significant impact on both the accuracy of decoding and the ability to identify true reading errors. This section contrasts Hybrid ASR [Povey et al. \(2011\)](#), Wav2vec2 [Baevski et al. \(2020\)](#), and Whisper [Radford et al. \(2022\)](#) models from the perspective of their suitability for miscue detection tasks.

2.3.1 Hybrid ASR Systems

Hybrid ASR systems traditionally consist of separate components: an acoustic model (AM), a language model (LM), and a pronunciation model with lexicon. These components are independently trained and later combined during decoding. For reading assessment tasks, particularly those involving prompted text (e.g., canonical stories), hybrid systems offer precise control over decoding behavior.

One of the strengths of hybrid ASR in reading applications lies in its language model constraints. By training the LM on the canonical reading material, the system can be biased to produce expected transcriptions, which helps in aligning the decoded text to the reference for WCPM scoring. However, this tight coupling with the LM makes hybrid systems less effective at detecting actual miscues [Gothi et al. \(2024\)](#). Substitutions and mispronunciations may go unnoticed if the LM overly influences decoding to favor grammatically and semantically likely outputs.

2.3.2 End-to-End ASR with Wav2vec2

In contrast, Wav2vec2 is an end-to-end (E2E) model based on self-supervised learning (SSL). Wav2vec2 learns speech representations without relying on transcribed text or a predefined vocabulary during pretraining. This absence of language-specific constraints allows it to capture low-level acoustic and phonetic details [Mohamed et al. \(2022\)](#), which is essential for identifying fine-grained miscues in children’s speech.

Wav2vec2 models can be fine-tuned either at the grapheme level or phoneme level. The model is primarily driven by its acoustic modeling capabilities. Although

Wav2vec2 occasionally produces incorrect spellings during decoding, researchers often integrate an external language model (LM) to improve transcription accuracy. However, for the specific task of reading miscue detection, using only the acoustic model (AM) is more beneficial. This is because we aim to capture actual misreadings or mispronunciations without the LM “correcting” them based on grammatical or lexical expectations [Gothi et al. \(2024\)](#).

Our objective is to build a system that relies primarily on acoustic features, with minimal influence from language modeling. Wav2vec2 supports this goal effectively—it learns strong acoustic representations during pretraining without any explicit language supervision. Interestingly, it also develops a degree of implicit language modeling capability, derived from patterns in the finetuning data. This balance allows Wav2vec2 to function as a robust AM that can detect miscues while remaining less biased by linguistic constraints, making it particularly well-suited for ORF assessment tasks.

2.3.3 Whisper and its Limitations

The Whisper model, introduced by OpenAI [Radford et al. \(2022\)](#), represents another end-to-end architecture, but it differs significantly from Wav2vec2. Unlike SSL-based models, Whisper is trained in a Large-Scale Weak Supervision fashion, language-aware paired audio-text datasets. It operates in an autoregressive manner—generating output tokens conditioned on previously predicted tokens—which introduces strong dependencies on linguistic patterns.

While Whisper achieves high transcription accuracy in general-purpose ASR, its grapheme-level modeling and strong LM influence make it less ideal for reading miscue detection. It often “corrects” incorrect pronunciations based on context,

leading to lower recall of miscues [Gao et al. \(2024\)](#). This behavior is desirable for general ASR applications but counterproductive for educational assessments, where the goal is to detect every deviation from the prompted text.

Empirical studies by [Gao et al. \(2024\)](#) have shown that Wav2vec2 exhibits higher recall for reading miscues compared to Whisper, making it more suitable for this application.

2.3.4 Hybrid + Wav2vec2: Leveraging the Best of Both

In our study, we explore a two-pass system that combines the strengths of both architectures [Gothi et al. \(2024\)](#):

- Hybrid ASR is used in the first pass to obtain accurate word-level alignments and time boundaries. This is particularly useful for downstream tasks such as calculating confidence scores and aligning miscues temporally.
- Wav2vec2, in the second pass, is used as a pure acoustic model, decoding without external language model influence. This pass helps capture true miscues that may have been ignored or corrected by the hybrid model.

A natural question arises: *Why not use Wav2vec2 in the first pass?* We observed that Wav2vec2 lacks precise word- or phone-level time alignment, which is critical for miscue localization. Its language model integration is also weaker compared to hybrid ASR, making it less effective when decoding constrained story text. Moreover, Wav2vec2 tends to drop disfluent or noisy words, whereas hybrid ASR, aided by its lexicon and LM, often recovers plausible hypotheses. Hence, we rely on hybrid ASR first, followed by Wav2vec2 for error-sensitive acoustic decoding.

Therefore, this dual-system approach allows us to exploit the temporal precision and language model bias of hybrid ASR in the first pass, while leveraging the acoustic robustness and LM-free decoding of Wav2vec2 in the second pass. Together, they enable a more comprehensive and accurate miscue detection pipeline. Further details on the system integration and experimental results will be presented subsequent chapters.

TABLE 2.1: Comparison of ASR Architectures for Reading Miscue Detection.

Model	Hybrid ASR Systems	Wav2vec2	Whisper
Training Approach	AM, LM, and pronunciation lexicon trained separately	SSL-based pretraining followed by fine-tuning	Trained with weak supervision on audio-text pairs
Decoding Behavior	Strong LM bias; favors expected transcriptions	Works better without LM to preserve errors	Autoregressive; high LM influence
Suitability for Miscue Detection	Good for alignment; may miss miscues due to LM correction	Captures acoustic/phonetic details well; robust AM	Tends to "correct" errors; low recall for miscues
Strengths	Accurate word-level alignments	Minimal linguistic bias; identifies subtle miscues	High transcription accuracy
Limitation	Over-corrects due to LM bias	Poor time resolution; drops disfluent words	Over-corrects based on LM context

2.4 Relevant Literature

This section reviews key research directions that enhance the use of Wav2vec2 in end-to-end (E2E) speech recognition, with a focus on phone vs. grapheme modeling, children’s speech adaptation, multilingual fine-tuning, pseudo labeling, and confidence feature extraction.

2.4.1 Phone vs. Grapheme Recognition in E2E ASR

End-to-end ASR models, such as Wav2vec2, directly map input audio to text without relying on intermediate components like a lexicon or pronunciation dictionary. During fine-tuning for ASR, a Connectionist Temporal Classification (CTC) head is

typically added on top of the Wav2vec2 encoder. This head predicts a sequence of discrete output tokens, the choice of which significantly affects model performance and interpretability.

In most ASR tasks, these output tokens are graphemes—characters or subwords corresponding to the target language’s script. Where Phone-based models represent speech using phonetic units, which are more closely tied to the actual acoustic properties of speech. This is particularly valuable in scenarios where the spoken form deviates significantly from the canonical text. For tasks like reading miscue detection, students often pronounce words in ways that do not correspond to any valid vocabulary item in the language. In such cases, it becomes necessary to transcribe the utterance using a phonetic script such as IPA or Devanagiri rather than English characters. For instance, the word “shepherd” in a Grade 3 text may be pronounced as “seperd” or “sheep hard”, among other variations [Gothi et al. \(2024\)](#). While some of these may result from L1 accent influences, many are phonetically driven misreadings. As such, phone-based models are more effective than grapheme-based ones for capturing and analyzing these deviations from canonical pronunciation.

In multilingual ASR, researchers have explored both grapheme-based and phone-based approaches. Grapheme-based models rely on distinct alphabets for each language, while phone-based models unify the phoneme inventory across languages. Studies such as [Żelasko et al. \(2020\)](#); [Feng et al. \(2021\)](#) demonstrate that mapping phones from multiple languages into a shared phonetic space improves cross-lingual transferability and benefits languages with limited labeled resources. Given that miscue detection aims to identify any acoustic deviation, including incorrect pronunciations, phone-based models are naturally more robust. They are less constrained by orthographic conventions and better aligned with the acoustic signal. This makes

phone-level modeling with Wav2vec2 especially suited to applications in reading assessment, where children’s speech often includes disfluencies and substitutions.

2.4.2 Wav2vec2 for Mispronunciation Detection

[Xu et al. \(2021\)](#) present one of the first studies to explore self-supervised learning for mispronunciation detection (MD) using Wav2vec 2.0. Traditionally, MD in Computer-Assisted Pronunciation Training (CAPT) systems [Yan et al. \(2020\)](#) has relied heavily on likelihood-based features such as Goodness of Pronunciation (GOP) and Log Phone Posterior (LPP), computed using forced alignments from hybrid ASR models [Witt and Young \(2000\)](#); [Hu et al. \(2015\)](#); [Franco et al. \(1999\)](#). In contrast, [Xu et al. \(2021\)](#) leverage large-scale unlabeled speech data for self-supervised pretraining with Wav2vec2 and fine-tune the model using a small amount of pronunciation-labeled data. They frame MD as a binary classification task, where for each segment aligned to a canonical phone, the model predicts whether the pronunciation is correct. The model takes as input the canonical phone sequence, alignment information, and waveform, and outputs phone-level correctness labels. Their results demonstrate the efficacy of Wav2vec2 for mispronunciation detection, particularly in low-resource or L2 learner contexts. [Peng et al. \(2021\)](#) conducted a study exploring the fine-tuning of Wav2vec2.0 models for mispronunciation detection and diagnosis (MDD). Their experiments demonstrated that self-supervised pretraining, followed by fine-tuning on limited labeled data, could achieve competitive F1-scores, highlighting the potential of Wav2vec2.0 in MDD tasks.

Building on this direction, [Ryu et al. \(2023\)](#) propose a multi-task learning framework that jointly optimizes Automatic Pronunciation Assessment (APA) and Mispronunciation Detection and Diagnosis (MDD). APA aims to automatically evaluate the

pronunciation quality of spoken language learners, typically by scoring aspects such as accuracy, fluency, and prosody. [Chen et al. \(2024\)](#). MDD focuses on identifying and localizing pronunciation errors at the phoneme or word level, providing targeted feedback for language learners [Shahin and Ahmed \(2024\)](#). Recognizing the strong correlation between phonetic errors and pronunciation scores in L2 speech, the authors design a unified architecture that uses CTC and cross-entropy losses to train both tasks concurrently. Their model is initialized with a pretrained Wav2vec2 encoder, which is first fine-tuned on phone recognition before joint training. Evaluated on the Speechocean762 benchmark, the joint model outperforms single-task systems and highlights the benefits of leveraging shared linguistic structure between pronunciation scoring and error detection. Recently [Gao et al. \(2025\)](#) investigated the further fine-tuning of Wav2vec2.0 models in low-resource settings to enhance children’s speech recognition and word-level reading diagnosis. Their study found that augmenting training data with similar out-of-domain speech improved model performance. The native child dialogue speech and non-native child reading speech are used as similar out-of-domain data for augmentation.

Inspired by these approaches, our method adapts Wav2vec2-based phone recognition for the task of reading miscue detection at the word level. While prior work has primarily focused on detecting phone-level correctness in second-language learners, our objective is to identify reading errors within the context of Oral Reading Fluency (ORF) assessments [Gothi et al. \(2024\)](#).

In our approach, we first fine-tune a Wav2vec2 model for phone recognition using a language-appropriate phoneme inventory. From the model’s frame-level predictions, we derive confidence scores that are aggregated across segments to estimate the reliability of each word. These aggregated scores form word-level confidence features, which are then used by a downstream classifier to label each word as either correct

or miscued. This method leverages the phonetic sensitivity of Wav2vec2 along with structured confidence estimation to enable robust word-level miscue detection—even in the presence of unconventional pronunciations commonly observed in children’s speech. The integration of this system with a hybrid ASR model to improve temporal alignment and feature reliability will be discussed in detail in a subsequent chapter.

2.4.3 Children’s Speech Adaptation and Pseudo-labeling

Our objective is to train an effective Wav2vec2-based phone recognition model using the best available data, with the end goal of applying it to reading miscue detection. Achieving high performance in this task involves addressing two major challenges: adapting to children’s speech and utilizing unlabelled audio through pseudo-labeling techniques.

A key challenge in training ASR systems for educational applications is the limited availability of labeled children’s speech data. While large-scale adult speech datasets are commonly available for many languages, child-specific datasets are much scarcer. This mismatch impacts performance, as acoustic and prosodic characteristics in children’s speech differ significantly from adults [Thienpondt and Demuynck \(2022\)](#).

This situation raises an important question: How can we build accurate child speech models despite limited resources? One promising direction is to explore cross-lingual transfer using multilingual ASR systems. For example, if labeled Marathi children’s speech is available, we investigate how it can help improve a Hindi ASR model. By sharing knowledge across languages during training, the model can better generalize to under-resourced languages and children speech.

Deep learning models typically require large amounts of labeled data, but manually annotating children’s speech is time-consuming and costly. To address this, we

explore pseudo-labeling, a method that allows us to make use of unlabeled audio by generating automatic labels during training [Lee \(2013\)](#).

Pseudo-label training has emerged as a promising approach for unsupervised model adaptation in low-resource settings. This technique relies on using a pretrained and fine-tuned model (not only SSL pretrained) to generate predictions (pseudo-labels) on unlabeled audio, which are then used as additional training data. A key aspect of this process is confidence estimation, which helps filter out unreliable predictions and retain only high-quality examples [Wang and Breckon \(2020\)](#); [Benato et al. \(2021\)](#); [Cascante-Bonilla et al. \(2021\)](#); [Xu et al. \(2020\)](#).

[Sivaraman et al. \(2022\)](#) proposed a method using CTC loss and probability ratio-based confidence (PRC) scores to select reliable unlabeled samples for training. Building on this idea, [Bhogale et al. \(2024\)](#) introduced multiple confidence estimators and a multimodal evaluation framework (SONAR) [Duquenne et al. \(2023\)](#), enabling large-scale pseudo-labeling for low-resource languages. Their work showed that this approach leads to significant improvements in ASR performance by effectively leveraging unlabelled speech.

In our work, we adopt similar strategies by combining pseudo-label generation with confidence-based filtering, enabling our model to learn from unlabeled children’s speech data and improve its ability to detect miscues even in low-resource scenarios.

2.4.4 Word-Level Confidence Estimation in End-to-End ASR

In automatic speech recognition (ASR), confidence estimation refers to predicting how likely a decoded output (e.g., word or token) is correct. This is especially important in reading miscue detection, where identifying potential recognition errors can guide error classification and downstream decisions [Gothi et al. \(2024\)](#).

While confidence estimation has been widely studied in hybrid ASR systems using decoding lattices and hand-crafted features, end-to-end (E2E) ASR systems pose new challenges. Unlike hybrid systems, E2E models—such as those based on Wav2vec2—lack explicit lattices or structured alternatives, making confidence scoring less straightforward. Additionally, E2E models typically operate on subword tokens (e.g., characters or byte-pair encodings), while many downstream tasks, including miscue detection, require confidence scores at the word level.

Recent work by [Oneață et al. \(2021\)](#) and [Laptev and Ginsburg \(2023\)](#) highlights key challenges in confidence estimation for end-to-end ASR: converting token-level scores into word-level confidence. They compute token probabilities, and apply techniques like negative entropy, temperature scaling, and dropout-based uncertainty to estimate confidence. These token scores are then aggregated (e.g., via sum pooling) to generate word-level confidences. The study shows that simple entropy-based methods with calibrated logits perform well, while dropout and ensemble techniques offer better accuracy at higher computational cost. The study emphasizes that even simple techniques like entropy-based confidence combined with temperature scaling provide strong confidence.

Additionally, some recent studies such as [Naowarat et al. \(2023\)](#); [Ravi et al. \(2024\)](#) have proposed using a separate transformer-based Confidence Estimation Model (CEM), which is trained specifically to predict word-level confidence scores—either as continuous values or binary labels—using curated datasets. While this approach can yield more calibrated predictions, it requires additional supervised training on confidence-labeled data.

We will describe the overall system architecture in Chapter 4, and discuss the details of Wav2vec2 training, pseudo-label generation, and word-level confidence feature extraction in the subsequent chapters.

Chapter 3

Datasets and Characteristics

In this chapter, we describe the datasets used for training the ASR acoustic models, as well as the test sets employed for evaluation. The training and evaluation data were sourced from multiple datasets, encompassing both adult and children’s speech across different languages. Table 3.1 provides an overview of these datasets, including their source, language, speaker type, speech style, and collection origin.

3.1 Sources

TABLE 3.1: Speech Datasets Sources and its characteristics

Source	Language(s)	Speaker Type	Speech Type	Collection Source
IITM	Hindi, English	Adult	Conversational & Read	IITM
AIS	Marathi	Children	Read	Vendor
WPP	English	Children	Read	WhatsApp
KV	Hindi, English	Children	Read	Tara App

In this work, the IITM dataset [IITM \(2021\)](#) is used as the adult speech corpus, while AIS Marathi, WPP English, and the KV dataset serve as children’s speech datasets.

We collected a substantial dataset via the Tara App as part of the nationwide Kendriya Vidyalaya (KV) Oral Reading Fluency (ORF) assessment. The recordings were gathered from students in Grades 3 to 8 across multiple phases and years.

In 2022, data collection focused on Grades 3 to 5 from the KVS Mumbai Region (covering Maharashtra and Goa), with recordings exclusively in English.

In 2023, additional data were collected for Grades 3 reading three stories in both Hindi and English.

In 2024, the dataset expanded further and was organized into three evaluation phases—Baseline, Midline, and Endline. Across Grades 3–8, each phase included three stories (two paragraphs each), yielding a total of 36 story paragraphs in both Hindi and English. From this 2024 dataset, we have only used Hindi data for our work.

KVS Data Subset selection

For English MPS Data : All the other raw audio recordings are first passed through the TDNN ASR with garbage model to get the decoded text. Using the alignment of the decoded text with the canonical text, the number of miscues (insertions, deletions and substitutions) of the child are detected. This is then normalized by the number of words spoken (insertions, corrections, substitutions) to get the normalized miscues count. Using a threshold on the normalized miscues value, it is determined if a recording is Ratable(<20%)/Transcribable(>20% and <80%)/WeakReader(>80%). We started transcription from the Ratable recordings. When initial data was transcribed and consolidated, we found that it was not representing examples with miscues so we added more recordings from the Transcribable category so that recordings with more miscues are represented.

For Hindi Data : we initially planned to use the data for phrasing assessment, so we initially considered only those student attempts from ASR results of the TDNN system which have 1) reading accuracy (i.e. percentage of correct words) of at least 95% across both paragraphs. 2) WCPMs of 85 and above average across both paras 3) Microphone connected since we wanted clean recordings. We had the same challenge here after initial data collection that we need to represent more data from lower proficiency children so we added more examples in the WCPM 70-85 category.

Further details on the test dataset are provided in the corresponding section.

3.2 Training Dataset

IITM English

For our application, we trained a baseline model using the IITM English ASR challenge, 2021 [IITM \(2021\)](#), which was released as part of the "Hindi-Tamil-English ASR Challenge." Each utterance in the dataset is a short segment from a larger set of recordings, consisting of one to three sentences lasting from 0.4 seconds (single word) to 15 seconds. Train set: 178 hours Val set: 5.3 hours This data consists of adult English reading and conversational speech.

WPP English

We have another manually transcribed children dataset in English. This dataset was obtained from the WPP India Foundation and involves recordings of students from grades 6 to 10, reading grade-appropriate paragraphs. The dataset consists of 22 distinct stories and 72 unique paragraphs, with two to four paragraphs for each

story. In total, there are 1104 speakers included in the WPP dataset. The average audio durations are about 33 seconds and containing about 70 words. The WPP dataset is partitioned into 6 folds based on story ID, speaker identity, and age group. For our experiments, we used 4 folds for training, 1 fold for development, and the remaining fold for testing results in WPP dataset.

IITM Hindi

The Hindi baseline model was trained using the IITM Hindi Challenge data, 2021 [IITM \(2021\)](#), which was released as part of the "Hindi-Tamil-English ASR Challenge." Each utterance in the dataset is a short segment from a larger set of recordings, consisting of one to three sentences lasting from 0.4 seconds to 15 seconds. Train set: 85 hours Val set: 22 hours This data consists of adult Hindi reading and conversational speech containing 2937 speakers.

AIS Marathi

The AIS Marathi dataset consists of recordings in the Marathi language, featuring 686 speakers aged between 7 to 21, reading age-appropriate sentences for a duration of 10 to 15 seconds. This dataset was collected by DAPLAB, and it provides a rich variety of voices from a wide age range. Since we wanted to adapt our Acoustic Model for children's speech, we used a subset of 303 speakers between the ages of 7-15.

KV Train - Hindi

We utilize a labelled training set compiled from the KV 2023 exercise and KV 2024 Baseline exercise.

We selected the following subsets for training:

- **KV 2023:** This subset includes 1,698 recordings from Grade 3 students, spanning a total of 18.41 hours and involving 1,219 unique speakers of Hindi.
- **KV Baseline 2024:** This includes 350 recordings from students in Grades 3 to 8, amounting to 4.61 hours of audio from 195 unique speakers.

Combined, the labelled training dataset comprises 2,048 recordings, **23.02 hours** of speech, and 1,414 unique speakers.

KV Unlabelled - Hindi

Recently, in 2024, we collected additional data from KV that had not yet been transcribed. To leverage this large pool of children’s speech, we incorporated it into training using a pseudo-labelling approach.

From this large unlabelled pool, we selected 6480 high-quality recordings based on the following criteria:

1. Automatic Speech Recognition (ASR) WCPM score ≥ 85 (Good reader)
2. Recordings captured using a microphone

3. Recordings sourced from the following 12 regions (Mainly from North India):

VARANASI, PATNA, AGRA, LUCKNOW, RANCHI, JAIPUR, GURUGRAM, DEHRADUN,
JABALPUR, BHOPAL, DELHI, MUMBAI

The total duration of selected recordings was:

- **Baseline:** 27.00 hours
- **Midline:** 26.74 hours
- **Endline:** 26.57 hours

Total Duration: 80.31 hours

To utilize these unlabelled recordings in model training, we applied a pseudo-labelling technique. The details of this approach and the selection process are discussed in Chapters 5 and 7.

TABLE 3.2: Summary of **English** language datasets used for training and evaluation.

Dataset	Utterances	Hours	Speakers
IITM (Train)	37223	185.00	1585
WPP (Train)	5674	62.24	1122
MPS (Test)	1600	19.00	1110

We computed the number of unique tokens (words) present in the ground truth transcriptions for each dataset. The IITM English dataset contains 23,013 unique tokens, while IITM Hindi contains 13,417 unique tokens. The WPP English children’s speech dataset includes 2,294 unique tokens, and the AIS Marathi children’s speech dataset has 303 unique tokens. For the KV Hindi children’s speech data, the KV 2023 subset contains 2,292 unique tokens, and the KV Baseline 2024 subset contains 3,134 unique tokens. It is important to note that pseudo-labeled data is in

TABLE 3.3: Summary of **Hindi-Marathi datasets** with utterance, hour, and speaker counts used for training and testing.

Source	Training Dataset			Pseudo-Labelled			KV 2000 (Test)			KV 812 (Test)		
	Uttr	Hrs	Spkr	Uttr	Hrs	Spkr	Uttr	Hrs	Spkr	Uttr	Hrs	Spkr
IITM Hindi	27131	107.00	2937	0	0	0	0	0	0	0	0	0
AIS Marathi	7835	7.21	303	0	0	0	0	0	0	0	0	0
KV 2023	1698	18.41	1219	0	0	0	1899	20.8	1252	0	0	0
KV Baseline 2024	350	4.61	195	949	12.05	942	101	1.58	63	0	0	0
KV Midline 2024	0	0	0	944	11.82	920	0	0	0	812	9.81	438
KV Endline 2024	0	0	0	971	12.12	971	0	0	0	0	0	0
Total	37014	137.23	4654	2684	36.00	2833	2000	22.38	1315	812	9.81	438

phone format and not in ground-truth word text, and was therefore excluded from this vocabulary analysis.

3.3 Test Datasets

MPS - English

As part of a benchmarking initiative aimed at evaluating reading levels in elementary education, audio recordings of oral reading in English (as a second language) were collected from approximately 2,500 students across 10 government schools in Maharashtra and Goa. This dataset came from one of the KV datasets. The data collection was conducted during the summer of 2022 and involved students from Grades 3, 4, and 5 (ages 7–11). Ethics approval was obtained for the recording process, with speaker identities anonymized, retaining only grade and gender information.

The students participating in this study are multilingual, typically exposed to both Hindi and English starting from Grade 1. In contrast to first-language (L1) English learners, these students possess little to no pre-existing vocabulary in English. As a

result, they encounter both the auditory and orthographic forms of English words simultaneously. This lack of aural familiarity, combined with the opaque orthography of English and the phonotactic constraints of their home languages, often leads to diverse types of pronunciation errors.

Each student was prompted to read two paragraphs from a single story appropriate to their grade level, with each paragraph containing approximately 60–80 words. A unique story was assigned to each grade. Word-level manual transcription is currently in progress. Observations from the transcribed data reveal a range of reading errors, including numerous non-English word substitutions, which are transcribed using phonemic notation.

For the purposes of this study, we selected a subset of the transcribed data comprising utterances in which at least 70% of the canonical words were attempted. This subset includes 1,600 utterances from 1,110 unique speakers, distributed roughly evenly across the three grade levels, and amounts to approximately 19 hours of audio. We refer to this subset as the **Maharashtra Primary Schools (MPS)** dataset. The audio quality varies, with background noise ranging from negligible to moderate classroom disturbances.

This dataset serves as the test set in our experiments. Ground-truth miscue labels—namely, Correct (Cor), Substitution (Sub), Deletion (Del), and Insertion (Ins)—are derived through a comparison of manual transcriptions with phonemically aligned canonical prompts, as detailed in [4](#).

Table [3.4](#) summarizes the distribution of attempted words and miscues across the dataset. On average, approximately 10% of attempted words are identified as miscues (i.e., Del + Sub), though the number of miscues per utterance varies between 0

TABLE 3.4: Distribution of word reading errors across the 1,10,898 attempted words in MPS dataset

Tags	Number (%)	Description
COR	99,554 (89.77%)	Prompt word is correctly pronounced
INS	3269 (2.94%)	Inserted word, not part of the prompt
DEL	1331 (1.20%)	Prompt word omitted
SUB-1	3452 (3.11%)	Substitution: one-phone difference
SUB-2	6561 (5.91%)	Gross substitution (>1 phone variation)

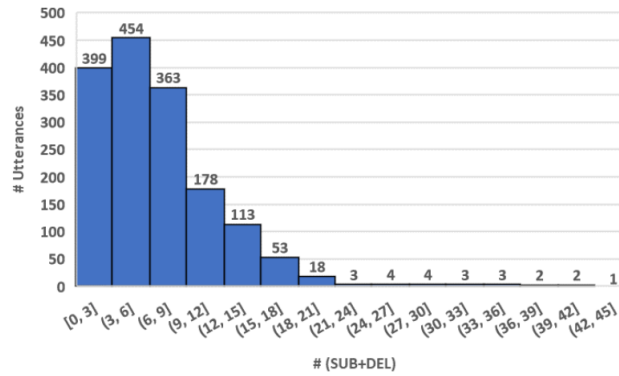


FIGURE 3.1: Distribution of % Miscue Rate for MPS dataset - English

and 24. The manual transcriptions provide fine-grained annotations that distinguish between different types of disfluencies.

To facilitate cross-validation (CV) in the training and evaluation of our miscue detection classifier (described in the second-pass system), the dataset is partitioned into six folds. These folds maintain non-overlapping speakers and a balanced distribution of grade levels and miscue densities.

KV 2000 - Hindi

Audio recordings of oral reading of Hindi grade-appropriate texts were collected as part of a large scale baseline reading assessment across a government school network in India for Grades 3 to 8 (ages 7-13) in 2023. Ethics clearance was obtained for the recordings, with speaker information anonymized except for grade and gender. The

students, who come from diverse linguistic backgrounds, begin learning to read and write in Hindi from Grade 1.

The text prompts comprise of the 2 paragraphs of a single story, each between 60-80 words. Manual transcription at the word level is underway. The reading errors are observed to comprise a number of non words, which are transcribed phonetically in Devanagari. Of the already transcribed dataset, we select for this work those utterances (i.e. the recording of one story paragraph) with a minimum level of expected fluency in terms of words correct per minute (WCPM) ≥ 70 . These amount to 2000 utterances across 1316 unique speakers, with 1900 recordings from grade 3 and 100 recordings from grade 4-8, for a total audio duration of 22 hours. We term this the KV 2000 Hindi dataset. The background noise varies from barely audible to classroom noise in the vicinity of the speaker. This dataset serves as our test dataset with ground-truth miscue labels (Cor/Sub/Del/Ins) derived by comparison of the manual transcription with suitably aligned text prompts (using the phonetically oriented text alignment discussed in [Gothi et al. \(2024\)](#))

The speakers are drawn from across the country with different first languages and we observe diverse reading errors and pronunciation variability due to regional accents and dialectal influences. Where applicable, we account for acceptable alternate pronunciations using a phoneme-based lexicon. Table 3.5 shows the distribution of attempted words across the dataset of 2000 prompts, where we note that about 9% of the total attempted words constitute reading miscues (i.e. DEL + SUB, including different types of SUB), while the per-utterance miscues range from 0 to 24 words. We observe a greater proportion of single-phone difference errors (expected to be challenging to detect automatically).

The test dataset is divided into 6 folds with non-overlapping speakers and uniform

TABLE 3.5: Distribution of word reading errors across the 1,28,955 attempted words in KV-2000 dataset

Tags	Number (%)	Description
COR	117,422 (91.1%)	Prompt word is correctly pronounced
INS	4629 (3.59%)	Inserted word, not part of the prompt
DEL	1729 (1.33%)	Prompt word omitted
SUB-1	9808 (4.65%)	Substitution: one-phone difference
SUB-2	3801 (2.94%)	Gross substitution (>1 phone variation)

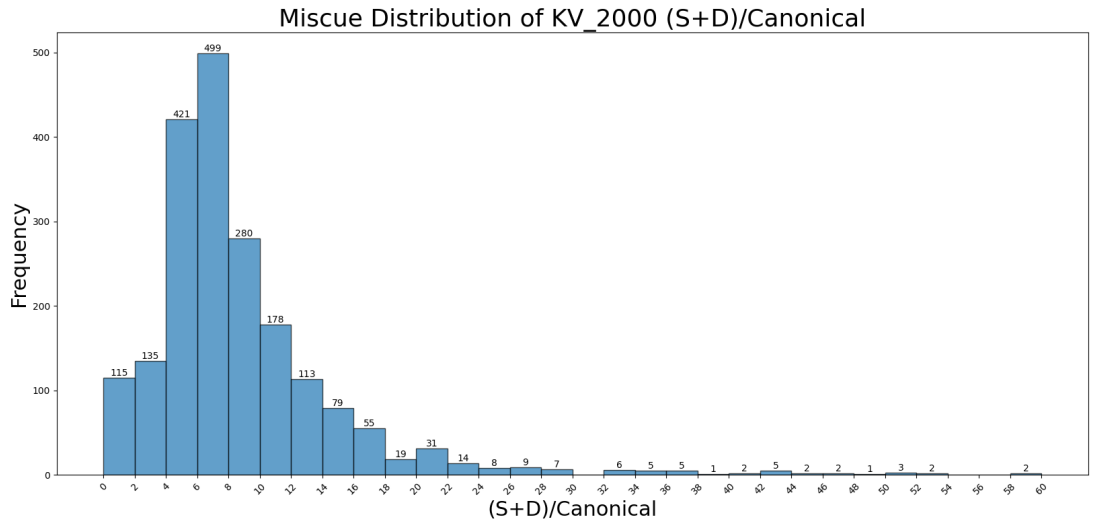


FIGURE 3.2: Distribution of % Miscue Rate for KV-2000

distribution of grades and reading levels (in terms of number of miscues per utterance) to facilitate the cross-validation (CV) training and testing of miscue detection classifier of the second pass. Figure 3.2 shows a histogram of % miscue rates across recordings.

KV 812 - Hindi

We have begun manual transcription of KV midline 2024 dataset. From the large pool of data, a subset of 812 recordings has been transcribed to date, covering a total of 9.81 hours of audio from 438 unique speakers, we called it as KV 812.

TABLE 3.6: Distribution of word reading errors across the 64427 attempted words in KV-812 dataset

Tags	Number (%)	Description
COR	60235 (93.49%)	Prompt word is correctly pronounced
INS	2293 (3.56%)	Inserted word, not part of the prompt
DEL	677(1.051%)	Prompt word omitted
SUB-1	1817(2.82%)	Substitution: one-phone difference
SUB-2	1698(2.63%)	Gross substitution (>1 phone variation)

Table 3.6 presents the distribution of attempted words in this subset. Approximately 6% of the attempted words are classified as reading miscues, defined here as the sum of Deletion (Del) and Substitution (Sub) errors, including phonetic variations of substitutions. Figure 3.3 illustrates the distribution of Miscue Rates across the dataset.

To support the evaluation of our miscue detection system in a robust and reproducible manner, this dataset has been partitioned into six cross-validation (CV) folds. The folds are constructed to ensure non-overlapping speakers and balanced distributions across grade levels and miscue frequencies per utterance, enabling fair training and testing of the second-pass miscue detection classifier.

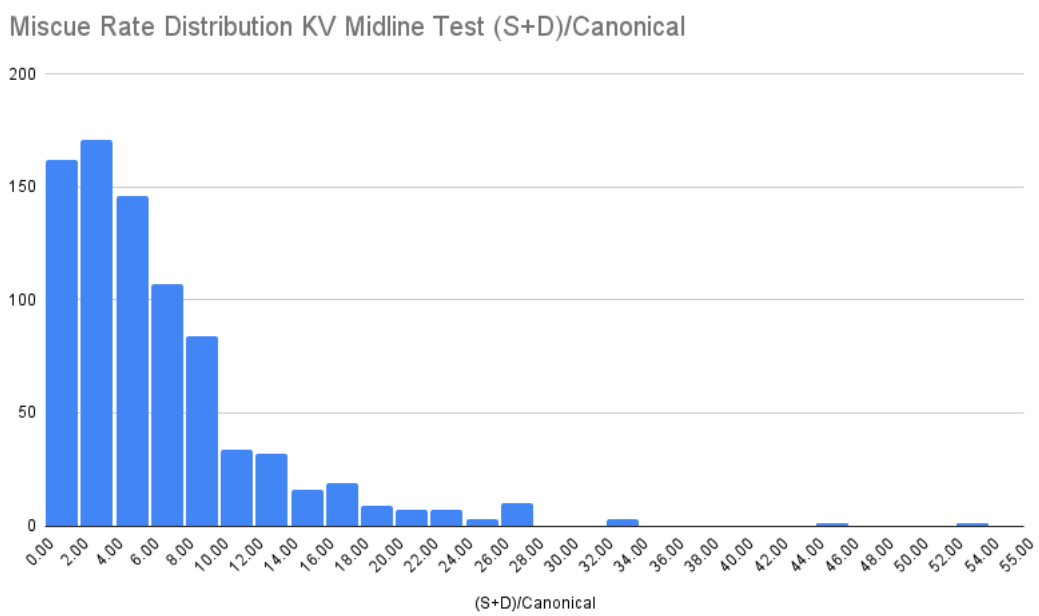


FIGURE 3.3: Distribution of % Miscue Rate for KV-812

Chapter 4

A two-pass miscue detection system

In this chapter, we provide a high-level overview of our proposed two-pass miscue detection system. The system is designed to combine the strengths of a strongly constrained hybrid ASR model with the acoustic sensitivity of Wav2vec2-based phone recognition, enabling robust and fine-grained detection of reading miscues. The process is divided into two passes:

In the first pass, the utterance is decoded using a hybrid ASR system. The resulting decoded text is aligned with the canonical reference text using a phonetic-aware alignment algorithm. This alignment outputs labels for each word as Correct (C), Substitution (S), Deletion (D), or Insertion (I). Words labeled as Substitution or Deletion are immediately marked as miscues. Words labeled as Correct (C), along with their time boundaries, are passed to the second pass for further analysis.

In the second pass, local word-level confidence features are extracted from both the hybrid ASR and the Wav2vec2 phone recognition model for each "C"-labeled word.

These features are then passed to a classifier, which makes a binary decision—determining whether a “C”-labeled word from the first pass is actually a true correct word or a missed substitution (miscue).

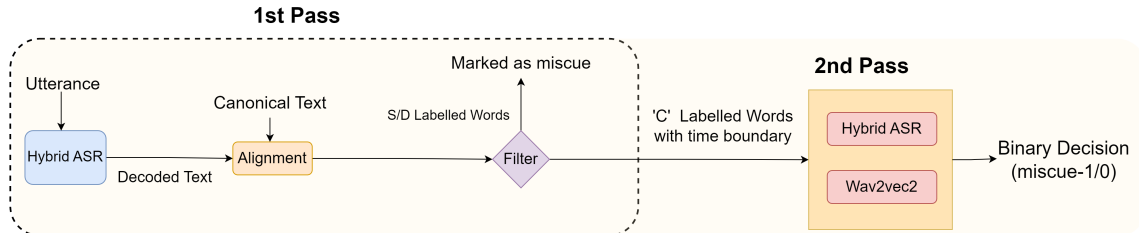


FIGURE 4.1: Overview of the proposed 2-pass miscue detection system.

4.1 First Pass: Hybrid ASR

The **first pass** of our system involves decoding the input utterance using a **hybrid ASR system**, which integrates a **TDNN-based acoustic model** and a **word-level trigram language model (LM)** trained on canonical reading prompts. The ASR output is aligned with the canonical text using a modified version of the alignment algorithm from [Ruiz and Federico \(2015\)](#). This alignment assigns each canonical word one of the following labels:

- **Correct (C)** – if it matches the decoded word,
- **Substituted (S)** – if the word was misread,
- **Deleted (D)** – if the word was skipped,
- **Insertion (I)** – if an extra word was added by the reader.

Additionally, the hybrid ASR provides **word-level time boundaries**, which are later used for confidence feature extraction.

We build this system for two languages: **English** and **Hindi**, and evaluate its performance using **Word Error Rate (WER)**, **Miss Rate (False Negative Rate)**, and **False Positive Rate (FPR)** on their respective test datasets.

To understand the impact of language model constraints, we compare two language models used in the first pass:

- (i) A **trigram LM trained on manual transcripts** of the test data (oracle case), and
- (ii) A **trigram LM trained only on canonical prompt texts** (practical deployment case).

In both cases, **language model interpolation** is applied. The LM trained on transcripts has prior knowledge of reading errors, providing an **upper bound on first pass performance**, while the prompt-trained LM reflects the realistic case where transcripts are not available at inference time.

English Language Results

TABLE 4.1: WER and miscue detection performance of Hybrid ASR with different LM constraints (**MPS English**)

Pass 1 LM	WER %	Miss Rate %	FPR %
Trigram on Transcripts	10.17	21	4.6
Trigram on Prompts	13.01	72	1.1

As seen in Table 4.1, using the LM trained on prompts results in a **3% absolute increase in WER** and a **significant rise in miss rate**. This indicates that many miscues are incorrectly decoded as canonical words, consistent with prior

observations [Molenaar et al. \(2023\)](#). Therefore, our goal is to **reduce the miss rate** from this prompt-based decoding, while keeping the FPR within acceptable limits at closer to LM Trigram on Transcripts.

Hindi Language Results

TABLE 4.2: WER and miscue detection performance of Hybrid ASR with different LM constraints (**KV 2000 - Hindi**)

Pass 1 LM	WER %	Miss Rate %	FPR %
Trigram on Transcripts	12.20	44.2	2.9
Trigram on Prompts	14.57	71.0	1.3

TABLE 4.3: WER and miscue detection performance of Hybrid ASR with different LM constraints (**KV 812 - Hindi**)

Pass 1 LM	WER %	Miss Rate %	FPR %
Trigram on Transcripts	13.50	23.70	6.40
Trigram on Prompts	16.61	55.10	2.32

For Hindi, we observe a similar pattern as in Table 4.2 and 4.3. The LM trained on transcripts yields a lower WER and miss rate compared to the prompt-trained LM. This is expected, as the transcript-trained LM can model the typical mistakes made by readers. However, in real-world applications, such transcript data is unavailable at inference time.

This high miss rate in the prompt-trained setting poses a key challenge: *how can we recover the miscues missed in the first pass?* To address this, we propose a **second-pass system** that operates on the “C”-labeled words from the first pass. This

system computes **confidence-based features** and uses them to make improved binary decisions about whether a word is truly correct or a miscue (i.e., an undetected substitution).

4.2 First pass : Post-Processing Alignment

In reading assessment tasks, aligning the ASR hypothesis with the canonical prompt text requires a more nuanced approach than conventional word-level Levenshtein alignment. Standard alignment methods treat words as discrete units and classify discrepancies as substitutions, insertions, or deletions. However, these methods fall short when dealing with complex phenomena in oral reading—such as word fragmentations, mispronunciations, false starts, or merged words—which are common in children’s speech [Gothi et al. \(2024\)](#).

To accurately identify such errors and to extract reliable time boundaries for downstream feature computation, we adopt a **phonetically informed alignment** strategy. This approach builds upon and modifies the alignment algorithm proposed by Ruiz et al. [Ruiz and Federico \(2015\)](#) to suit the specific needs of reading miscue detection.

The alignment process proceeds in two stages:

Stage 1: Initial Word-Level Alignment

The canonical text and the decoded ASR output are first aligned using standard **word-level Levenshtein distance**. Each word in the canonical sequence is tagged as: C/D/I/S

Stage 2: Phonetic Re-Alignment for Error Spans

When an **error span** is detected—a sequence of words between two Correct (C) labels containing S, D, or I tokens—we apply a refined alignment step:

1. The words in the error span from both canonical and ASR hypothesis sides are converted into **phoneme sequences** using a pronunciation lexicon with syllable boundary annotations.
2. Each phoneme is treated as an individual token. Special **boundary tokens** are inserted to mark syllable and word boundaries.
3. A **phoneme-level Levenshtein alignment** is applied to these sequences, using the following constraints:
 - Boundary tokens are not allowed to align with phonemes.
 - Vowel phonemes can only align with other vowels (including r-colored vowels, but not semivowels).
 - Consonant phonemes can only align with other consonants (including semivowels).
4. The alignment output is used to update the original word-level alignment tags, refining the classification of substitutions and allowing for improved handling of **word splitting or merging**.

In addition to phonetic similarity, the algorithm accounts for **multiple valid pronunciations** of a canonical word, as defined in the pronunciation lexicon. It also accommodates **false starts**, a common occurrence in children’s oral reading, by introducing a rightward alignment bias that avoids prematurely labeling partially spoken words as deletions or insertions.

This approach is inspired by the **Phonetically-Oriented Word Error Rate (POWER)** metric, which introduces structural constraints on phoneme alignments and better reflects phonetic similarity than simple string comparison.

Final Label Assignment

After phonetic re-alignment, each canonical word is associated with a corresponding decoded word (if not deleted). The final label for each word is determined as follows:

- If the decoded word matches the canonical word after phonetic re-alignment, it is labeled as **Correct (C)**.
- If a mismatch remains, the word is labeled as a **Substitution (S)**.
- Words in the prompt that are unmatched in the hypothesis are labeled as **Deleted (D)**.
- **Insertions (I)** from the hypothesis are ignored for scoring, consistent with conventions in oral reading assessment.

More practical related details of post processing is in lab github repo ([link](#)). This enhanced alignment strategy allows us to better identify reading miscues and produce high-quality input for the second-pass feature extraction and classification steps.

4.3 Second Pass: Confidence Feature Calculation

In the second pass, we focus on reevaluating the words initially labeled as **Correct (C)** in the first-pass alignment. Although these words are matched to the canonical prompt by the hybrid ASR, many miscues remain hidden due to the language

model’s strong constraints. To detect such errors, we compute a set of **word-level confidence features** using both the **Hybrid ASR system** and the **Wav2vec2-based phone recognition model**. These features serve as inputs to a classifier that decides whether each “C”-labeled word is indeed correct or a miscue (typically a substitution that escaped the first pass).

Hybrid ASR Confidence Features

- **Lattice-based Confidence Scores:** Derived from the word graph generated during decoding, these scores capture the dominance of the selected hypothesis relative to alternatives in the lattice [Jiang \(2005\)](#). We compute:
 - Word posterior probability,
 - Likelihood Ratio Test (LRT),
 - Hypothesis density.
- **Goodness of Pronunciation (GOP):** GOP assesses the acoustic similarity between the predicted and canonical phone sequences [Witt and Young \(2000\)](#); [Kanters et al. \(2009\)](#). We use Kaldi’s implementation to compute GOP scores from senone-level posterior probabilities [Hu et al. \(2015\)](#). The scores are aggregated at the word level using:
 - Mean across frames,
 - Mean and minimum across phones.

Wav2vec2-Based Confidence Features

For each “C”-labeled word, we extract its time-aligned acoustic segment and pass it through a fine-tuned **Wav2vec2 phone recognition model**. The model predicts

the corresponding phone sequence, which we compare to the canonical pronunciation as in [Gothi et al. \(2024\)](#):

- **Levenshtein Distance (LevDist)**,
- **Cost Levenshtein Distance (CostLevDist)** – using a weighted phone confusion matrix.

To further quantify prediction certainty, we compute additional **confidence scores** at the word level by applying: Temperature scaling, Negative entropy, Aggregation to combine token-level confidence into word-level scores.

These confidence features will be discussed in detail in Chapter 8 and techniques used for scaling and aggregation.

4.4 Feature Fusion and Classification

We combine the confidence features from both the Hybrid ASR and Wav2vec2 models and input them into a **binary classifier**. This classifier is trained on labeled reading data to learn discriminative patterns that distinguish subtle miscues from truly correct pronunciations.

The system’s performance is evaluated using **Detection Error Tradeoff (DET) curves** [Martin et al. \(1997\)](#) and **Area Under the Curve (AUC)** metrics. Given the imbalanced nature of miscue detection, we maintain a fixed **False Positive Rate (FPR)** threshold to ensure meaningful evaluation.

Chapter 5

Building Wav2vec2 Phone Recognition Model

In this chapter, we describe the process of building a phone recognition system using the Wav2vec2 model. We focus on acoustic model (AM) training by extending the architecture with a linear classification head. Later sections discuss tokenization strategies, Preprocessing dataset, loss function formulation, and decoding technique (at Test time). We also outline the generation of pseudo labels for unlabelled audio and the integration of a language model (LM) for decoding.

5.1 Acoustic Model (AM) Training

Detecting reading miscues requires reliable recognition of phone sequences spoken by the reader. One common way to measure the mismatch between a spoken word and its expected pronunciation is through the **Levenshtein distance** between their

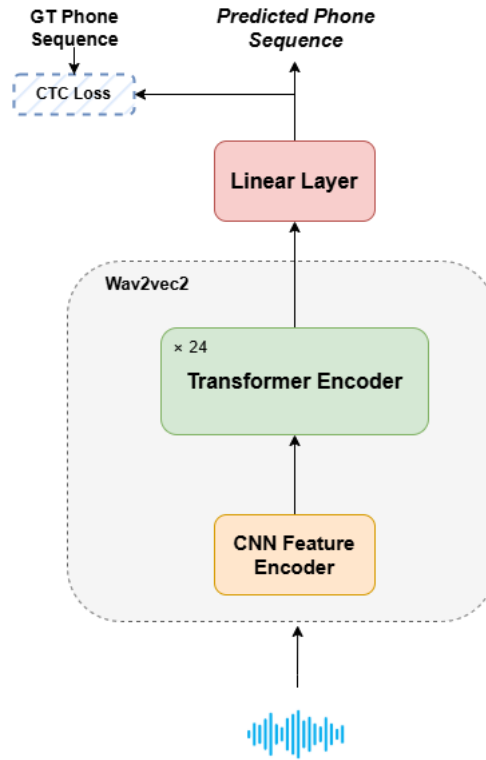


FIGURE 5.1: Wav2vec2-Large Phone Recognition model architecture

phone sequences. However, this requires an accurate phone recognizer—a task traditionally considered difficult, especially in multilingual and low-resource settings.

With the emergence of self-supervised learning (SSL) models such as Wav2vec2, it has become possible to learn high-quality acoustic representations from raw audio. These representations capture detailed phonetic information and have been shown to achieve strong performance on both within- and cross-language phone recognition tasks.

Wav2vec2 Fine-Tuning for Phone Recognition

We adopt the Wav2vec2 architecture as shown in Figure 5.1 . It consists of:

- A **convolutional feature extractor** that encodes the raw audio waveform into latent representations,
- A deep **Transformer encoder** that models long-range dependencies in speech,
- A **linear classification head** added on top, projecting the final encoder layer’s output (of size 1024) into a distribution over phone vocabulary tokens.

The model is fine-tuned on labeled datasets using phone-level sequences as target transcriptions instead of word-level text. These phone-level labels are derived from manual word transcriptions through forced alignment, which is necessary because each word may have multiple valid pronunciations, making simple lexicon lookup insufficient.

Forced alignment is performed using a hybrid ASR model, which provides accurate time-aligned phone transcriptions. Specifically, for Hindi, the hybrid model is trained on a combination of IIM Hindi and AIS Marathi datasets, while for English, a model trained on IITM English and WPP English is used.

To explicitly mark word boundaries in the resulting phone sequences, a special * token is inserted between the phone sequences of consecutive words. This is done by aligning phone-level CTM and word-level CTM time stamps.

Suppose the reference transcription is **”He counted the ship”**. After forced alignment, the corresponding training ground truth at the phone level might look like: **HH IY * K AW N T EH D * DH AH * SH IY P ***. Note that pronunciation variants may exist for certain words (e.g., “ship” could be realized with different vowel phones such as IH or IY), and the forced alignment process selects the most likely phone sequence based on the actual audio.

These phone sequences, enriched with boundary markers, serve as the ground truth targets for fine-tuning the model in both monolingual and multilingual training setups.

Tokenization and Vocabulary Design

We use **phone-based tokenization**, which supports better cross-language transfer and generalization compared to grapheme-based models. Two phone vocabularies were defined [11.1](#):

- For **English**, we use a set of 44 tokens (40 phones + special tokens) ,
- For **Hindi and multilingual training**, we use a superset vocabulary of 57 tokens (53 phones + special symbols).

The vocabulary includes a special token `*` to denote word boundaries, which helps group phones into word-level sequences—useful for downstream alignment and evaluation.

CTC Loss Function

To train the model, we use the **Connectionist Temporal Classification (CTC)** loss function. CTC is suitable for sequence-to-sequence tasks where input and output lengths are misaligned and where frame-level alignment is not known a priori.

CTC works by:

- Predicting a sequence of probability distributions over tokens (including a special **blank** symbol) for each input frame,

- Summing over all valid paths that collapse to the correct target sequence (using rules to merge repeated tokens and ignore blanks),
- Maximizing the log-probability of the correct target sequence across all valid alignments.

This formulation enables efficient training without needing explicit frame-to-phone alignments.

5.2 Inference and Phone Sequence Prediction

During inference, given an input audio segment, the trained Wav2vec2 model outputs a sequence of probability distributions (logits) over the phone vocabulary—one per frame. To derive the final phone sequence:

1. We select the token with the highest probability at each frame,
2. Remove repeated tokens and blank symbols using **CTC decoding** rules,

Example : Let's say wav2vec2 model has predicted below output by choosing highest probability token at each frame.

Frame-level output:

```
HH HH  €  €  €  IY  €  €  *  *  *  €  €  K  K  AW  €  N  N  €  €  T  T  €
EH EH  D  *  *  *  €  €  DH DH  €  AH  €  *  *  SH  €  IY  IY  €  P  P
*  *
```

Final Output:

HH IY * K AW N T EH D * DH AH * SH IY P *

(Note: Repeated tokens and blank symbols (ϵ) are removed to obtain the final output.)

Wav2vec2 processes audio using a fixed frame size of 20 ms. Therefore, a 2-second audio segment is divided into approximately 100 frames. In the frame-level output sequence generated by the model, the blank symbol typically dominates, appearing more frequently than language-specific tokens. This is a characteristic behavior of CTC-based models, where the blank token helps maintain alignment flexibility between input frames and output symbols.

The resulting sequence can then be compared with the canonical pronunciation to detect phone-level deviations.

5.3 Pseudo Label Generation (Unlabeled Data)

Deep learning models generally benefit from access to large quantities of labeled training data. However, in the context of Automatic Speech Recognition (ASR), there is a notable scarcity of labeled children’s speech data—particularly for low-resource languages. While some high-resource languages have a few hours of labeled children’s data, collecting and annotating similar data for underrepresented languages is both expensive and time-consuming.

At the same time, it is often feasible to collect large amounts of **unlabeled children’s speech**, as discussed in the dataset descriptions in Chapter 3, where the KV Unlabeled Hindi dataset was presented. In later chapters, we show that including even a small amount of labeled children’s data significantly improves model performance. This motivates us to explore how we can leverage **unlabeled speech data** using a technique called **pseudo labeling**.

Pseudo labeling involves using a pre-trained model to generate transcript hypotheses for unlabeled audio. These hypotheses are then filtered using confidence scores to ensure only reliable examples are included in further training. This strategy enables the model to generalize better across speech from diverse speakers and domains without requiring additional manual transcription.

Unsupervised Adaptation using Confidence Filtering

We follow the unsupervised adaptation framework proposed by [Sivaraman et al. \(2022\)](#), which employs two confidence metrics to filter pseudo-labeled data:

- Connectionist Temporal Classification (CTC) loss
- Probability Ratio Confidence (PRC) score

CTC Loss as Confidence Score

In a CTC-based ASR system, the model is trained to map an input sequence of acoustic frames $X = [X_1, X_2, \dots, X_T]$ to a target sequence $Y = [Y_1, Y_2, \dots, Y_N]$ of output tokens (e.g., phonemes or characters). The CTC loss for a training pair (X, Y) is defined as the negative log-likelihood of the target sequence given the input:

$$\text{CTC}_{\text{loss}} = -\log p(Y|X)$$

During inference, this loss can also be calculated using the **model's own prediction** $\hat{Y}$. The resulting value serves as a measure of model **confidence**: lower CTC loss implies higher confidence in the output transcript. Importantly, CTC loss is naturally normalized for utterance duration and output length, making it suitable for comparing across different utterances.

Probability Ratio Confidence (PRC) Score

In beam search decoding, the ASR model generates a ranked list of output hypotheses $\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n$, each with an associated conditional probability $p_i = p(\hat{Y}_i|X)$. Since the hypotheses are ranked by likelihood, we always have $p_1 > p_2 > \dots > p_n$. The Probability Ratio Confidence (PRC) score is defined as:

$$\text{PRC} = 1 - \frac{p_2}{p_1}$$

Although the ASR model produces frame-level logits as output, beam search decoding uses these logits to explore multiple likely token sequences. At each decoding step, beam search maintains a list of top-scoring partial hypotheses, ultimately generating a ranked list of complete sequences $\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n$, each with an associated score reflecting its likelihood.

The PRC score captures the difference between the top two competing hypotheses, offering insight into the model's confidence. It correlates with transcription quality and can be used as a filtering criterion.

Filtering and Dataset Preparation

To build a pseudo-labeled dataset, we:

1. Use a baseline fine-tuned Wav2vec2 phone recognition model to decode unlabeled children's speech,
2. Compute the CTC loss and PRC score for each utterance,

3. Apply thresholding on the desired score to retain only high-confidence examples.

We apply this method to the KV Unlabelled - Hindi dataset, where large amounts of unlabeled children’s speech are available. The resulting pseudo-labeled dataset is then used to further fine-tune our phone recognition model, improving performance on the child speech domain.

We will present the results of this process, including the impact on the finally achieved phone error rate (PER), in Chapter 7.

5.4 Language Model Integration

In this section, we describe our experiments with integrating a language model (LM) into the Wav2vec2-based acoustic model (AM). While this component is not directly related to the second-pass miscue detection pipeline, it represents our initial exploration of using Wav2vec2 in a first-pass ASR setting—specifically as a grapheme (character) recognition model.

Our early hypothesis was that a fine-tuned Wav2vec2 model, when combined with a strong LM, could serve as a standalone recognizer in the first pass of the reading assessment pipeline. However, as we describe later, several limitations of this approach led us to instead use Wav2vec2 primarily in the second pass. This section outlines the LM integration experiments we performed using a character-level Wav2vec2 model.

Decoding with Language Model

To decode the output emissions from the Wav2vec2 model, we use a lexicon and a separately trained word-level n-gram language model, built using the canonical story text (reading prompts) and trained as a 3-gram LM with Kneser-Ney smoothing using the KenLM toolkit. This LM provides task-specific language constraints during decoding.

Let $y = y_1 y_2 \dots y_M$ be a candidate word sequence. Let $p_{AM}(y)$ and $p_{LM}(y)$ denote the scores from the acoustic model and the language model, respectively. The optimal sequence is obtained using the following scoring function:

$$y^* = \arg \max_y [\log p_{AM}(y) + \alpha \cdot \log p_{LM}(y) + \beta \cdot |y|] \quad (5.1)$$

Here, α controls the weight of the language model, β is the word insertion bonus, and $|y|$ is the length of the hypothesis. Decoding is performed using an efficient beam search algorithm as described in [Pratap et al. \(2019\)](#); [Javed et al. \(2022\)](#).

Hyperparameter Tuning and Beam Search

We conducted our experiments on the **KV Hindi November 2023** dataset, which consists of approximately 7 hours of labeled data.

Initially, we performed a coarse grid search over a wide range of hyperparameters:

- $\alpha \in [-4, 4]$ (language model weight)
- $\beta \in [0, 5]$ (word insertion bonus)

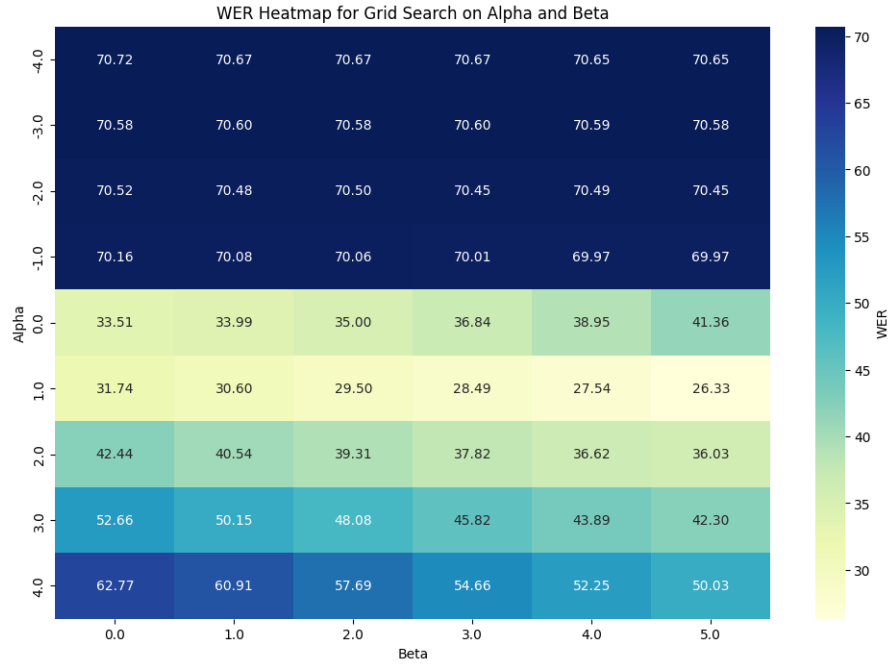


FIGURE 5.2: WER (%) heatmap from initial grid search over a broad range of hyperparameters: $\alpha \in [-4, 4]$, $\beta \in [0, 5]$. Significant variation observed with small parameter changes.

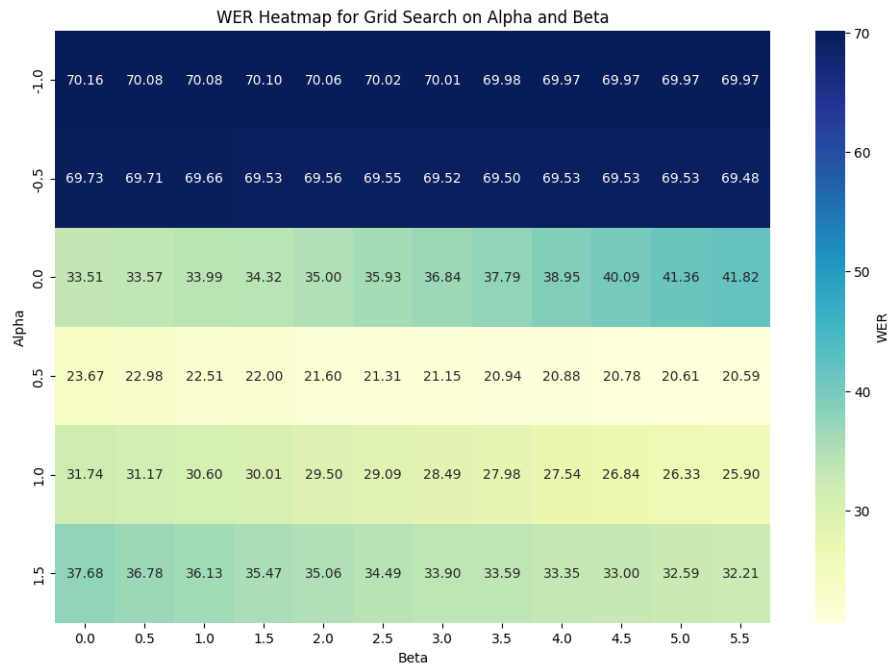


FIGURE 5.3: WER (%) heatmap from refined search: $\alpha \in [-1.5, 1.5]$, $\beta \in [0, 7.5]$ (step = 0.5). Best results clustered around $\alpha = 0.5$.

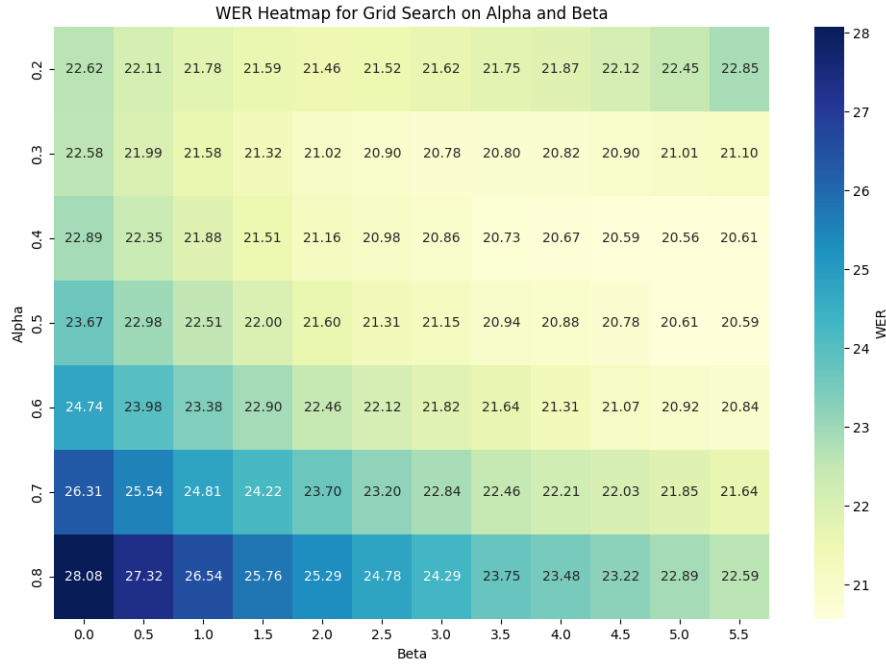


FIGURE 5.4: Fine-grained heatmap showing best WER of 20.56% at $\alpha = 0.4$, $\beta = 5.0$. Used for final beam size experiment.

Figure 5.2 presents a heatmap of Word Error Rate (WER) values across different combinations of α and β . The results show that even small changes (e.g., ± 1) in either parameter can lead to significant differences in WER.

Based on these observations, we refined our search:

- $\alpha \in [-1.5, 1.5]$ (step size = 0.5)
- $\beta \in [0, 7.5]$ (step size = 0.5)

The second heatmap (Figure 5.3) shows that optimal WER values cluster around $\alpha = 0.5$. The best result in this range was a WER of **20.56%**, achieved at $\alpha = 0.4$ and $\beta = 5.0$ (Figure 5.4).

Finally, using this optimal configuration, we increased the beam size to 1024 to enable a wider search space during decoding, which further reduced the WER to **20.33%** on the KV Hindi November dataset.

Model Setup and Evaluation

We used the IndicWav2vec2 Hindi model as the base and fine-tuned it further on IITM Hindi and AIS Marathi datasets for grapheme-level prediction. The models were evaluated on two datasets—KV Hindi November (7 hours) and KV Hindi December (22 hours)—using LM decoding with canonical text prompts.

TABLE 5.1: WER (%) Comparison Across Model Configurations for Two Hindi Datasets

Model Configuration (WER %)	KV Hi Dec 2023 (22h)	KV Hi Nov 2023 (7h)
Hybrid ASR + LM (Canonical)	7.86	16.00
W2v AM_IITM + LM (Canonical)	11.79	21.39
W2v AM_IITM_AIS + LM (Canonical)	11.88	20.33

From Table 5.1, we observe that the Hybrid ASR system outperforms Wav2vec2 across both datasets. This is expected, as the Hybrid ASR model is tightly constrained by a language model trained specifically on canonical story text. Its search space is narrow and optimized for high-confidence decoding of known word sequences.

In contrast, the Wav2vec2+LM system is more flexible and may occasionally decode words outside the lexicon or introduce spelling errors. In some cases, Wav2vec2 failed to detect certain words entirely, despite them being present in the audio. This was particularly noticeable in the KV Hindi November dataset, which includes a wide

range of speaker types, including less fluent readers and noisy recordings. We also observed some mismatches due to phonetic similarity and regional accents, especially in samples from South Indian students, which sometimes led to incorrect manual transcripts.

These results motivated our decision to use Wav2vec2 primarily in the second pass of our miscue detection system, where its frame-level acoustic sensitivity is more useful.

Chapter 6

English Language – Wav2vec2 Model

In this chapter, we present the development and evaluation of a Wav2vec2-based phone recognition model for English, using a combination of adult and children’s speech datasets. We compare its performance with a hybrid ASR system, using Phone Error Rate (PER) as the primary evaluation metric. The phone tokenization scheme used in this work is based on the English phone set described in [Appendix 11.1](#)

6.1 Model Training Setup

We begin by comparing the PER of the hybrid and Wav2vec2 systems under similar training data conditions. While the hybrid system is trained from scratch,

Wav2vec2 is initialized with a self-supervised pretrained model (XLSR) and subsequently fine-tuned using labeled data. Both models are evaluated using a phone-level transcription on the same test set.

The Wav2vec2 model is fine-tuned using the **Connectionist Temporal Classification (CTC)** loss function. The convolutional feature encoder is frozen during fine-tuning, while all Transformer encoder layers are updated. A linear classification head is added on top to map the output features to phone tokens. (We are also experimenting with unfreezing the convolutional feature encoder to evaluate its potential impact.)

Training Procedure

Fine-tuning the model on adult speech data helps it learn the Hindi language more effectively due to the larger volume of available data. Subsequently, we fine-tune the model on children’s speech to adapt it to the target domain, which is children’s speech recognition.

Fine-tuning is performed in two stages:

- Stage 1: 2 epochs on the IITM dataset (178 hours of adult speech),
- Stage 2: 5 additional epochs on the WPP dataset (40 hours of children’s speech).

For both fine-tuning setups, the convolutional feature encoder was frozen, while all Transformer encoder layers and the linear output layer were trained.

We use the **AdamW optimizer** [Loshchilov and Hutter \(2017\)](#) with:

- Learning rate: 1×10^{-4} (IITM), 1×10^{-5} (WPP fine-tuning),
- Dropout [Srivastava et al. \(2014\)](#): 0.1,
- Warmup steps: 200 (applied during IITM fine-tuning),
- Batch size: 16,
- Weight decay: 5×10^{-4} .

Several combinations of training configurations were explored to achieve the best performance. Initially, various learning rates and training durations were tested on the IITM dataset. The most effective setup involved starting from the SSL-pretrained model with a learning rate of 1e-4, followed by fine-tuning on children's speech data using a reduced learning rate of 1e-5 to retain the linguistic knowledge acquired from adult Hindi speech. For regularization, standard dropout and weight decay values recommended in the literature were experimented with, and the best-performing configuration was selected based on validation performance.

The hybrid ASR system uses a TDNN acoustic model trained on IITM and WPP, with a phone bigram language model trained on WPP transcripts.

6.2 Results and Analysis

TABLE 6.1: Phone Error Rate (PER) on the MPS English test set for different ASR systems and training strategies. The "Combined" setting refers to training on both IITM and WPP datasets from scratch using the SSL model. The "Fine-tuning" setting indicates continued training of a model first trained on IITM data and then fine-tuned on WPP data. TL: Transformer Layer, Conv: Feature Convolution layer

Model (Training Data)	PER (%)
Hybrid (IITM AM, WPP LM)	24.73
Hybrid (IITM $\rightarrow$ WPP AM, WPP LM)	18.15
Wav2vec2-XLSR (IITM only)	23.24
Wav2vec2-XLSR (IITM + WPP combined)	12.13
Wav2vec2-XLSR (IITM $\rightarrow$ WPP (only TL) fine-tuning)	9.67
Wav2vec2-XLSR (IITM $\rightarrow$ WPP (Conv+TL) fine-tuning)	9.72

Table 6.1 presents the Phone Error Rate (PER) on the MPS English test set. Wav2vec2 achieves significantly better performance than the hybrid ASR system when trained on the same data. In particular, we observe a sharp reduction in PER after fine-tuning on the WPP children’s dataset, improving from 23.24% to 9.67%.

This result confirms that **self-supervised pretraining enables Wav2vec2 to learn richer, more discriminative representations**, which can be effectively adapted to downstream tasks with limited labeled data.

Impact of Children’s Data

Fine-tuning on children’s speech (WPP dataset) led to a substantial reduction in PER for Wav2vec2, highlighting its sensitivity to domain-specific acoustic variation. This result suggests that:

- Pretrained models initially fine-tuned on adult speech tend to underperform on children’s speech due to domain mismatch,
- Including even a small amount of in-domain (children) data significantly boosts performance,
- Wav2vec2 benefits more from domain adaptation than traditional hybrid systems.

This observation motivated us to explore similar domain adaptation strategies in Hindi, including the use of **pseudo-labeled children’s data**, which will be discussed in Chapter [7](#).

Additionally, qualitative inspection of decoded phone sequences from Wav2vec2 showed close correspondence with manual transcripts. Most errors were localized to short phone spans, sometimes replaced entirely by blank symbols. The hybrid system, by contrast, showed broader mismatches likely due to language model constraints and acoustic mismatches.

Chapter 7

Hindi language - Wav2vec2

Multilingual Model

In this chapter, we present the development of a Wav2vec2-based phone recognition model for Hindi using a multilingual fine-tuning strategy. We incorporate both adult and children’s speech datasets—across languages—along with pseudo-labeled data to enhance model performance, particularly for children’s speech. The results are benchmarked using Phone Error Rate (PER) and compared against a hybrid ASR system.

Given the scarcity of labeled children’s speech, especially in low-resource languages like Hindi, we adopt a multilingual approach with a shared phoneme vocabulary. We use a superset of 52 phones to enable cross-lingual transfer as described in the appendix. This phone-based modeling, unlike grapheme-based systems, generalizes better across languages and allows the use of labeled other Indian language children’s data to benefit target language training.

In the following sections, we describe:

- Multilingual fine-tuning results using labeled data and different fine-tuning strategies,
- Pseudo label generation and filtering for Hindi children’s speech, and its experimental evaluation,
- Experiment of adding Hindi children’s labeled data (i.e., the target language children’s speech).
- Final results and comparative analysis of Wav2vec2 and hybrid ASR models on the Hindi test set.

7.1 Multilingual Fine-Tuning Experiments

As shown in Table 3.3, the Hindi training data comprised multiple datasets, including IITM Hindi (adult speech), AIS Marathi (children’s speech), and others.

We began our experiments with the XLSR Wav2vec2 SSL model, first fine-tuning it on the IITM Hindi dataset (adult speech). We then explored additional fine-tuning strategies using the AIS Marathi dataset (7 hours, children’s speech). These included partial fine-tuning of different subsets of Transformer layers (top, middle, bottom) and low learning rate adaptation. We also tested combining IITM Hindi and AIS Marathi data prior to fine-tuning, rather than fine-tuning sequentially on one dataset after the other.

Additionally, we experimented with including the WPP English children’s dataset (40 hours) to assess if domain adaptation from other unrelated languages would help. However, this led to worse performance, likely due to phonotactic mismatch between Hindi and English.

Training hyperparameters mirrored those used in the English experiments: all Transformer layers were trained while keeping the convolutional encoder frozen (unless stated otherwise) and 5 epochs.

We use the **AdamW optimizer** [Loshchilov and Hutter \(2017\)](#) with:

- epoch : 5 (For all experiments)
- Learning rate: 1×10^{-4} (IITM Hindi / Combined approach) and 1×10^{-5} if further fine-tuning on Children data (AIS/KV Train dataset),
- Dropout [Srivastava et al. \(2014\)](#): 0.1,
- Warmup steps: 200,
- Batch size: 16,
- Weight decay: 5×10^{-4} .

TABLE 7.1: PER (%) on KV-2000 Hindi using XLSR-based Wav2vec2 model fine-tuned with various strategies. Combining IITM and AIS datasets yields better performance than sequential fine-tuning. Where arrow $\rightarrow$ shows further fine-tuning and + shows combining the dataset. TL: Transformer

Model	PER (%) on KV 2000
XLSR IITM	10.16
XLSR IITM $\rightarrow$ AIS (All TL Layer)	10.30
XLSR IITM $\rightarrow$ AIS (Bottom)	10.27
XLSR IITM $\rightarrow$ AIS (Middle)	10.34
XLSR IITM $\rightarrow$ AIS (Top)	10.38
XLSR IITM + AIS (Combined)	10.02
XLSR IITM + AIS + WPP (Combined)	13.25

From Table 7.1, we observe that combining datasets yields better results than sequential fine-tuning. Therefore, all subsequent experiments used combined training datasets.

In the next phase, we evaluated the AI4Bharat IndicWav2vec2 model. Despite only marginal gains, it consistently outperformed XLSR. Given that IndicWav2vec2 was pretrained on more Indian language data, it likely captured phonetic features of the target languages more effectively.

TABLE 7.2: PER (%) on KV -2000 Hindi comparing XLSR and AI4Bharat IndicWav2vec2 models under various training configurations. Fine-tuning the convolutional layer (Conv) along with Transformer layers significantly improves performance on children’s speech.

Model	PER (%) on KV - 2000
XLSR IITM	10.16
AI4B IITM	10.12
XLSR IITM + AIS (Combined)	10.02
AI4B IITM + AIS (Combined)	9.98
AI4B IITM + AIS (Combined, +Conv layer finetuned)	9.67

Table 7.2 shows that fine-tuning the convolutional layer in addition to the Transformer layers provides a substantial improvement. Contrary to the recommendation in the original Wav2vec2 paper to keep the conv layer frozen, we found that updating it for children’s speech data yields significant PER gains. This adaptation appears crucial for learning the acoustic variability characteristic of children’s speech.

Based on these findings, we chose the AI4Bharat SSL model with full layer adaptation (Conv + Transformer) as our setup for subsequent pseudo-labeling experiments.

7.2 Pseudo Label Generation, Filtering and Experiments

In this section, we describe the process of generating and filtering pseudo-labeled Hindi children’s speech data to enhance model performance. Two filtering strategies are employed: one based on Connectionist Temporal Classification (CTC) loss

and the other on Probability Ratio-based Confidence (PRC). These filtered pseudo-labeled datasets are then combined with the existing labeled datasets (IITM Hindi and AIS Marathi) for further fine-tuning. We also compare the performance of models trained using these enhanced datasets.

Generating and Filtering Data

To address the limited availability of labeled Hindi children’s speech data, we leverage a pseudo-labeling approach. Specifically, we augment our training data using the KVS Unlabeled Hindi Dataset (**Baseline and Midline 2024 Experiment**), which consists of 54 hours of children’s speech audio collected during baseline and midline oral reading assessments. This dataset comprises 4,320 audio samples across 72 unique paragraphs.

We perform inference on this dataset using our previously trained model (AI4B SSL fine-tuned on IITM Hindi and AIS Marathi), and then apply utterance-level confidence scoring to filter high-quality samples. Following the method proposed in [Sivaraman et al. \(2022\)](#), we compute two types of confidence scores:

- **CTC Loss:** Calculated between the predicted label sequence (used as pseudo-ground truth) and the model’s own frame-level predictions.
- **Probability Ratio Confidence (PRC):** Derived from the ratio of probabilities between the top two beam search candidates as described in [5](#).

Using these metrics, we curate two distinct subsets of pseudo-labeled data, each approximately 22 hours in duration. These subsets are selected based on diversity across stories and thresholding the confidence metrics.

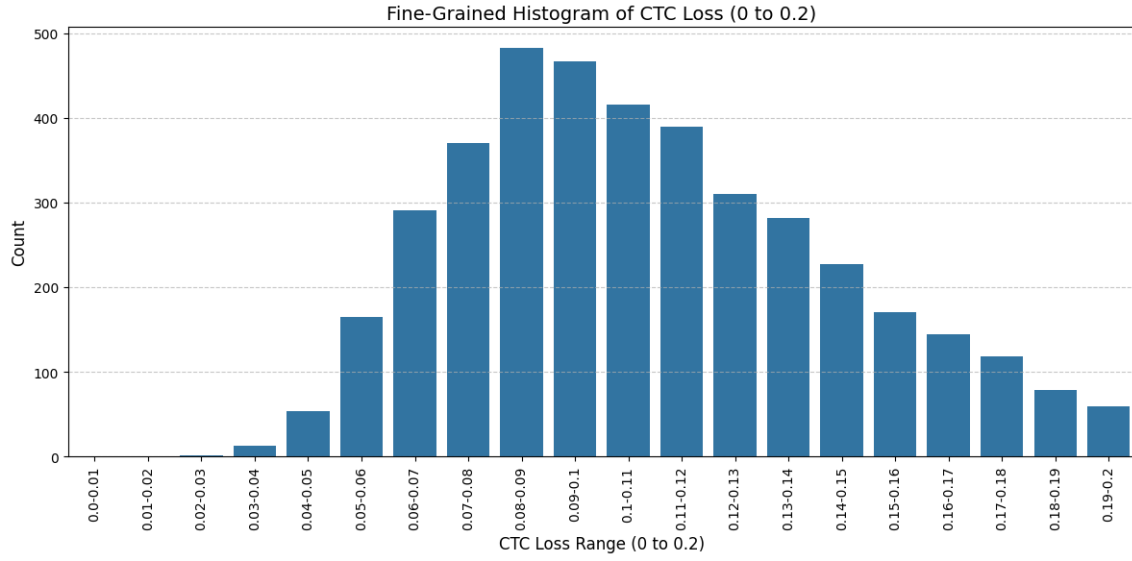


FIGURE 7.1: Distribution of CTC Loss values across the unlabelled Hindi children’s dataset. Most utterances lie around 0.1.

As shown in Figure 7.1, the CTC loss values are heavily skewed toward 0, with most falling below 0.1. We selected a threshold of 0.11 to filter utterances, resulting in a pseudo-labeled subset of approximately 22 hours with a diverse set of stories.

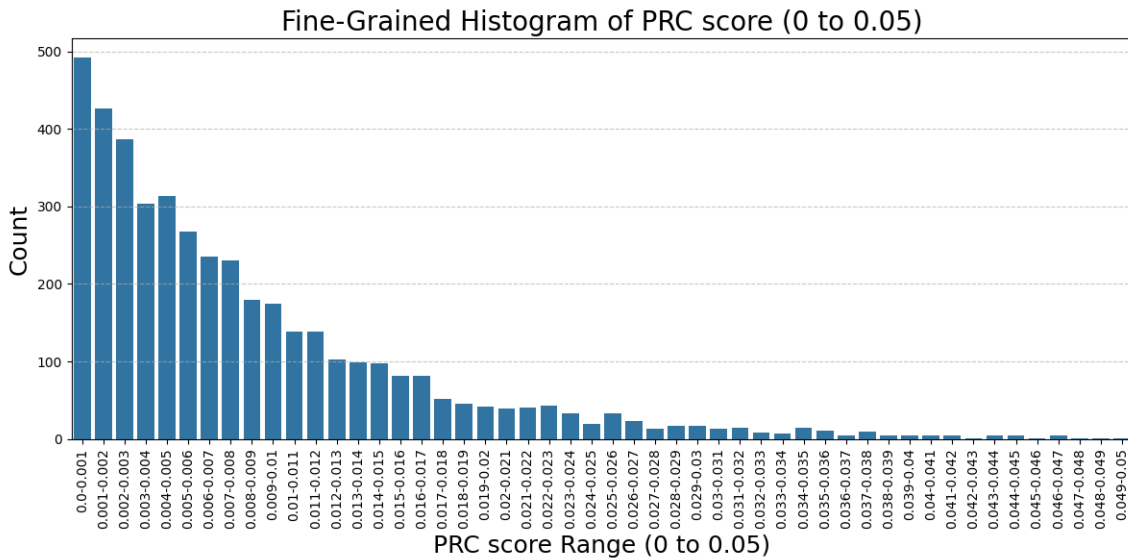


FIGURE 7.2: Distribution of PRC scores from beam search decoding across the unlabelled Hindi children’s dataset. Most scores are concentrated near zero.

For the PRC-based selection, we follow the process outlined in Chapter 5, using the top two beam search hypotheses to compute the PRC score. As seen in Figure 7.2, most scores are concentrated on the lower end. A threshold of 0.005 was chosen to select utterances, again yielding about 22 hours of diverse and high-confidence pseudo-labeled speech.

Threshold Selection

To validate our thresholds for filtering pseudo-labeled data, we conducted a study on KV Train dataset. For each utterance, we computed the corresponding **CTC Loss** and **PRC score**, then measured the **Phoneme Error Rate (PER)** against these metrics.

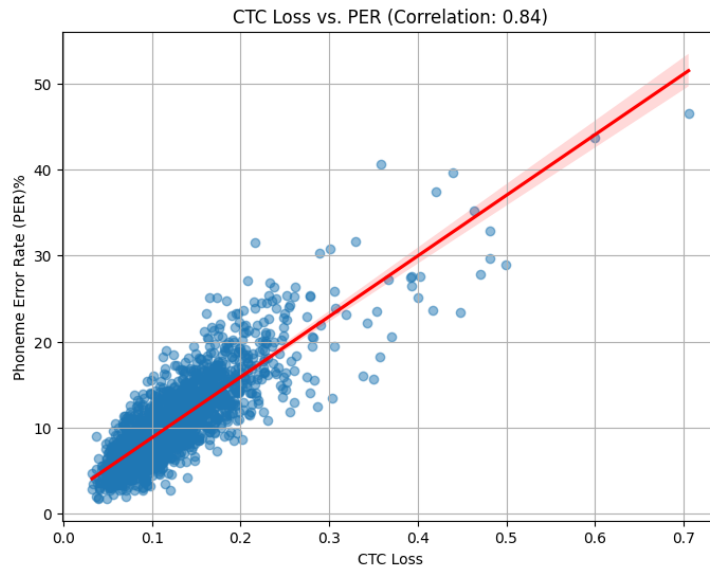


FIGURE 7.3: CTC Loss vs. PER. A strong positive correlation (0.84) indicates CTC loss is a reliable predictor of transcription quality.

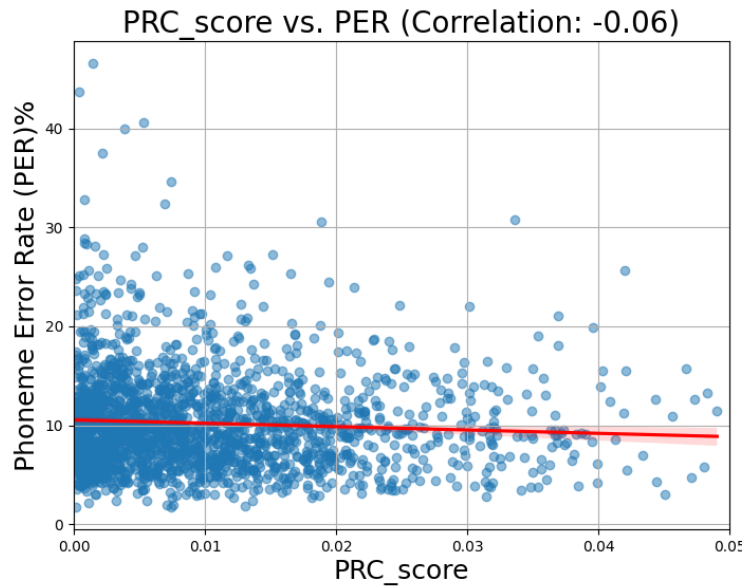


FIGURE 7.4: PRC Score vs. PER. A weaker negative correlation (-0.08) is observed.

As shown in Figure 7.3, **CTC loss exhibits a strong correlation with PER**, validating its effectiveness for ranking utterances by confidence. A CTC loss threshold of **0.11** captures high-quality predictions while excluding low-quality outputs such as noisy, silent, or distorted speech. Most utterances below this threshold have a PER less than 15, indicating that the model’s predictions are relatively accurate. Therefore, we apply this threshold to select pseudo-labeled data for training, which aids in adapting the model to children’s speech in the target language.

In contrast, the PRC score in Figure 7.4 shows a weaker correlation with PER. We select a threshold of **0.005** for PRC, identifying confident examples for use as pseudo-labeled training data. While a few outliers with higher PER still appear below this threshold, they are relatively sparse, and the majority of selected utterances fall within a PER range of 0 to 20.

Based on these insights, we use:

- **CTC Loss** < 0.11 for confident utterances
- **PRC Score** < 0.005 for confident utterances

These thresholds balance high confidence with data diversity and are empirically validated using labeled data.

Low-Confidence Utterance Selection for Manual Transcription

Beyond data augmentation, confidence scoring is also critical for selecting utterances that require **manual transcription**. By identifying low-confidence predictions, we can prioritize them for human review and correction, ensuring that the most uncertain or noisy inputs are handled with care.

We developed an end-to-end **Confidence Estimation and Audio Selection Pipeline** to route utterances accordingly. This enables targeted transcription of ambiguous samples, which improves model robustness when re-integrated into training.

Two filtering strategies are used:

1. CTC Loss (Efficient to compute)

- Utterances with **CTC Loss** > 0.11 are considered low confidence.
- Extremely high values (>1.0) often correspond to silent or malformed audio.
- To curate samples for transcription, we stratify selections by:
 - $0.11--0.20$, $0.20--0.30$, $0.30--0.40$, ..., >0.50

- Higher CTC loss ranges are sampled more heavily due to greater likelihood of acoustic errors (e.g., background noise, mic issues, Wav2Vec2 decoding gaps).

2. PRC Score (Beam-search dependent, slower)

- Utterances with **PRC Score** > 0.005 are treated as less confident.
- Scores > 0.06 typically indicate silence or very short audio.
- Transcription selection follows a similar stratified strategy:
 - $0.005--0.01$, $0.01--0.02$, $0.02--0.03$, ..., >0.03
- No single score band maps cleanly to specific error types, but high PRC ranges often correspond to degraded signal quality or decoding uncertainty.

By selecting a range of low-confidence utterances, this pipeline improves data quality, promotes diversity, model generalization and maximizes the impact of manual labeling efforts.

In the next subsection, we will present and discuss the experimental results based on these two filtered datasets.

Experiments

From Table 7.3, we observe that adding pseudo-labeled data to the IITM Hindi baseline reduces PER significantly: from 10.12% to 8.69% and 8.59% using CTC and PRC filtering respectively. PRC-based filtering yields slightly better results. Further improvements are achieved when AIS Marathi data is also included. The best result of 8.40% PER is obtained by combining IITM, AIS, and PRC-filtered

Model	PER (%) on KV 2000
XLSR IITM	10.16
AI4B IITM	10.12
AI4B SSL IITM + AIS	9.67
AI4B SSL IITM + Pseudo_labelled_CTC	8.69
AI4B SSL IITM + Pseudo_labelled_PRC	8.59
AI4B SSL IITM + AIS + Pseudo_labelled_CTC	8.49
AI4B SSL IITM + AIS + Pseudo_labelled_PRC	8.40

TABLE 7.3: PER (%) on KV 2000 - Hindi comparing baseline models with those trained using pseudo-labeled data (Baseline and Midline - 2025) filtered by CTC and PRC. PRC-based filtering yields the best performance.

pseudo-labeled data. This improvement is attributed to the combination of labeled children’s speech from AIS Marathi and pseudo-labeled children’s speech in the target language.

Ablation Study

To further evaluate the effect of pseudo-labeled children’s speech data, we conducted an ablation study across progressively larger subsets of the KV Hindi dataset. We trained models using:

- Only baseline pseudo-labeled data,
- Baseline plus midline pseudo-labeled data,
- Baseline, midline, and newly added endline-2025 unlabelled audio (pseudo-labeled).

These configurations help isolate the contribution of each data segment toward performance improvement. All experiments used the AI4Bharat SSL model, fine-tuned

with labeled adult Hindi (IITM), labeled Marathi children’s data (AIS), and pseudo-labeled Hindi children’s speech filtered using the PRC method.

Model Configuration	PER (%) on KV - 2000
XLSR IITM	10.16
AI4B SSL IITM	10.12
AI4B SSL IITM + AIS	9.67
AI4B SSL IITM + AIS	
+ Pseudo_labelled_PRC (B, 12 hrs)	8.77
+ Pseudo_labelled_PRC (B + M, 23 hrs)	8.40
+ Pseudo_labelled_PRC (B + M + E, 34 hrs)	8.28

TABLE 7.4: Ablation results using PRC-filtered pseudo-labeled data. B: Baseline (12.05 hrs), M: Midline (11.82 hrs), E: Endline (12.12 hrs) of 2024-25 assessment pseudo labelled data. Adding more diverse and high-quality children’s speech improves PER.

As shown in Table 7.4, each additional set of pseudo-labeled data contributes to a consistent improvement in performance. The best PER of 8.28% was achieved using the full training configuration that includes all labeled and pseudo-labeled subsets. This confirms the effectiveness of high-confidence pseudo-labeling and highlights the importance of continuously expanding labeled training resources—especially for children’s speech in low-resource languages.

7.3 Experiment of Adding Hindi Labelled Data of Children

In this section, we present the impact of incorporating labeled Hindi speech data from children into our training pipeline and evaluate its effectiveness on two different test sets: one with overlapping stories and one with non-overlapping stories.

As discussed in Chapter 3, we recently transcribed the **KV Baseline-2023** and **KV Baseline-2024** datasets, contributing over 23 hours of labeled Hindi speech from students in Grades 3–8. This combined dataset is referred to as **KV Train Hindi**. Previous experiments have already indicated that child-specific data significantly improves PER.

We evaluate the model performance on two types of test sets:

- **KV 2000 (Overlapping Stories):** This test set contains stories that also appear in the KV Train Hindi as described in chapter 3. This setting simulates a practical scenario where we want to use a fixed set of reference stories for a given grade across academic years. Since the model has seen the transcribed data of the same stories during training, it is expected to perform better at detecting miscues typical to those stories.
- **KV 812 (Non-overlapping Stories):** This set includes entirely new stories which are part of the KV midline 2024 exercise and not present in the KV Train Hindi dataset. It reflects real-world deployment, where new stories are introduced during testing. Success on this test set indicates the model’s generalization capability.

To explore the best strategy for utilizing the labeled KV Train Hindi data, we consider two training approaches:

1. **Fine-tuning:** Starting from an already trained model, we fine-tune it further using the KV Train Hindi data.
2. **Combined Training:** We include the KV Train data along with existing training data and retrain the model from scratch.

Model	KV 2000	KV 812
XLSR IITM	10.16	9.85
AI4B SSL IITM	10.12	9.80
AI4B SSL IITM + AIS	9.67	7.86
AI4B SSL IITM + AIS		
+ Pseudolabelled_PRC(B)	8.77	7.31
+ Pseudolabelled_PRC (B+M)	8.40	6.95
+ Pseudolabelled_PRC (B+M+E)	8.28	6.88
AI4B SSL IITM + AIS + Pseudolabelled_PRC (B+M+E)		
→ KV Train Hindi (Fine-tuned)	4.45	6.74
+ KV Train Hindi (Combined)	5.39	6.62
AI4B SSL IITM + AIS		
→ KV Train Hindi (Fine-tuned)	4.73	7.17
+ KV Train Hindi (Combined)	5.55	6.88

TABLE 7.5: PER (%) comparison of different models on KV2000 (overlapping) and KV-812 (non-overlapping) Hindi test sets. B: Baseline (12.05 hrs), M: Midline (11.82 hrs), E: Endline (12.12 hrs) of 2024-25 assessment pseudo labelled data and KV Train Hindi dataset of 23 hours.

Table 7.5 summarizes the performance of different configurations. We observe the following trends:

- We observe that the baseline model (XLSR IITM) achieves a slightly lower Phone Error Rate (PER) on the KV 812 dataset compared to the KV 2000 dataset. Upon examining the characteristics of the datasets, we found that the KV 812 dataset includes a higher proportion of good readers. As a result, there is a lower likelihood of pronunciation or reading errors in this dataset, leading to better performance (lower PER) for the baseline model.
- Gradual incorporation of pseudo-labeled data from Baseline (B), Midline (M), and Endline (E) assessments results in consistent improvement on both test

sets, with up to 1.5% absolute gain in PER for the KV-812 set.

- When we add the **KV Train data** on top of the best-performing model (AI4B SSL IITM + AIS + Pseudolabelled_PRC (B+M+E)):
 - **Fine-tuning** yields a significant improvement on **KV 2000**, reducing PER from 8.28% to **4.45%**, thanks to the overlap between test and training stories.
 - On the **KV-812** set, where no such overlap exists, the **combined training** approach performs slightly better (6.62% PER vs. 6.74% with fine-tuning), indicating better generalization to unseen texts.
- Additionally, we conducted experiments where the KV Train Hindi data was added **without** any pseudo-labeled data. The results show that:
 - Pseudo-labeled data is highly valuable when labeled data is limited.
 - The performance achieved by using more pseudo-labeled data is comparable to that of using labeled data alone, as seen in the KV 812 results (6.88% PER). This reinforces the effectiveness of self-supervised pseudo-labeling, especially in low-resource scenarios.

In summary, these experiments validate the importance of leveraging child-specific, language-specific labeled data. Fine-tuning is beneficial when the testing stories overlap with training, while combining labeled data in full retraining better supports generalization to new, unseen content.

7.4 Comparison of Wav2Vec2 and Hybrid ASR

In this section, we compare the performance of the Hybrid ASR system and the Wav2Vec2-based model across different training configurations and test datasets.

Table 7.6 presents the Phone Error Rate (PER) on two test sets: **KV 2000**, and **KV 812**. From the results, we observe that the Wav2Vec2-based models consistently outperform the Hybrid ASR system across all configurations. This is primarily due to Wav2Vec2’s stronger acoustic modeling capabilities and the advantage of leveraging large-scale self-supervised pretraining. While the Hybrid ASR approach benefits from explicit linguistic constraints, it falls short in generalizing to diverse or unseen story content, especially when trained on limited labeled data.

Model	KV 2000	KV 812
Hybrid ASR (AM, Phone-bigram LM (NCERT Text))		
IITM	19.48	20.19
+ AIS	13.95	12.15
+ Pseudolabelled_PRC(B)	-	-
+ KV Train Hindi	10.4	10.04
Wav2Vec2 Model		
AI4B SSL IITM	10.12	9.80
AI4B SSL IITM + AIS	9.67	7.86
+ Pseudolabelled_PRC (B+M+E)	8.28	6.88
→ KV Train Hindi (Fine-tuned)	4.45	6.74
+ KV Train Hindi (Combined)	5.39	6.62

TABLE 7.6: PER (%) comparison of Hybrid ASR and Wav2Vec2 models on KV 2000 (overlapping) and KV 812 (non-overlapping) Hindi test sets. B: Baseline (12.05 hrs), M: Midline (11.82 hrs), E: Endline (12.12 hrs) pseudo-labeled data. KV Labeled Hindi dataset: 23 hours.

Conclusion

Based on the extensive experiments conducted in this chapter, we summarize the key findings as follows:

1. **Adding child-specific data** significantly improves model performance by enabling the model to learn speech patterns unique to children. This adaptation to the target domain is crucial for Oral Reading Fluency (ORF) tasks.
2. **Combining the training set and fine-tuning both the convolutional feature encoder and the transformer layers** of the Wav2Vec2 model leads to a greater reduction in PER compared to only fine-tuning the transformer layers. Since the target domain—children’s speech—is acoustically different, fine-tuning the convolutional layers helps the model better adapt to these low-level characteristics.
3. **Gradual addition of pseudo-labeled data** (Baseline, Midline, Endline) from the target language contributes to steady improvements, particularly in low-resource settings.
4. When a labeled Hindi children’s dataset is available, **fine-tuning on it drastically reduces PER** for overlapping stories (KV 2000). For unseen story scenarios (KV 812), however, **combining the labeled dataset with the rest of the training data** is more effective, enabling the model to generalize better.
5. **Both pseudo-labeled data and a limited amount of labeled data** in the target language contribute significantly to performance improvements. Their combination also provides benefits, particularly in low-resource scenarios.

6. Finally, Finally, **Wav2Vec2 significantly outperforms Hybrid ASR** across all configurations, establishing it as the more suitable architecture for the phone recognition task. This is particularly useful in the second pass to extract word-level confidence features.

These insights guide us toward building more effective and scalable speech assessment systems, especially for under-resourced languages and populations.

Chapter 8

Confidence Feature

Once we have developed a robust phone recognition model, we use it during the second pass of our system to compute word-level confidence features.

During the first pass, many substitution errors (miscues) are missed. Therefore, all words labeled as "Correct" in the first pass are passed to the second pass. These words, along with their start and end time boundaries (provided by the hybrid ASR system), are analyzed using the Wav2vec2 model. Since Wav2vec2 does not provide precise word-level alignments, we rely on the hybrid ASR's timestamps.

In this chapter, we describe the calculation of Levenshtein-based edit distance features (LevDist), cost-based distance (CostLevDist), and word-level E2E confidence features derived from the Wav2vec2 output distribution.

8.1 Levenshtein Distance: LevDist and CostLevDist

For each aligned word from the hybrid ASR, we obtain its time boundaries and map them to Wav2vec2 frame indices (note: hybrid ASR has a frame size of 10 ms while

Wav2vec2 uses 20 ms).

We extract phone sequences from the given frame range using Wav2vec2 outputs. Since Wav2vec2 predictions include blank symbols and repeated tokens, we apply a CTC decoding algorithm to clean up the predictions.

From the lexicon file, The corresponding canonical word’s (from the reading prompt) phone sequence is extracted, which includes multiple pronunciations. We then compute the edit distance between the canonical phone sequence and the predicted phone sequence from Wav2vec2—this forms our LevDist feature as in figure 8.1.

To refine this further, we introduce a cost-based edit distance (CostLevDist), where phonetic substitutions with high confusion likelihood (e.g., vowels or similar-sounding phones) are given lower penalties. The confusion matrix between phones is computed using the development dataset. The substitution cost is calculated as:

$$\text{cost}[i, j] = 1 - \text{confusion_score}[i, j]$$

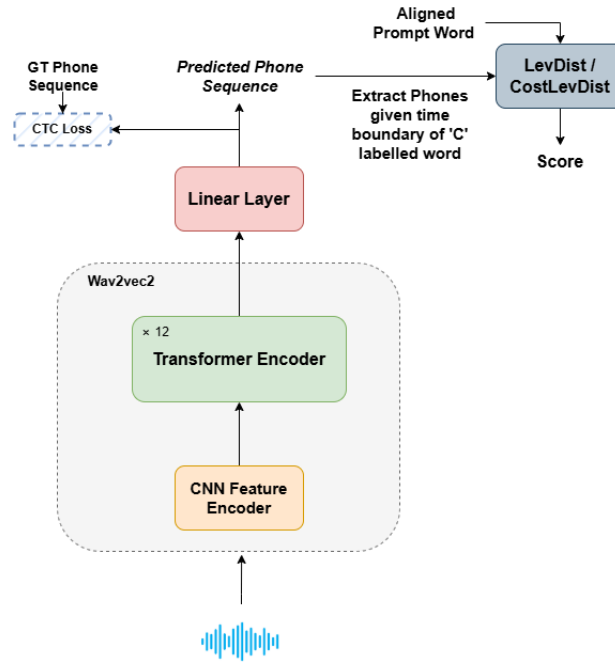


FIGURE 8.1: Phone sequence extraction and Levenshtein distance computation using Wav2vec2 outputs and hybrid ASR alignment.

8.2 End-to-End Confidence Features

The confidence score of the word should be high if model is confident about its prediction and low if it is not confident. Once we obtain the token-level posterior probability vector p_k for each time step from Wav2vec2, we calculate word-level confidence scores in two stages as in Figure 8.2:

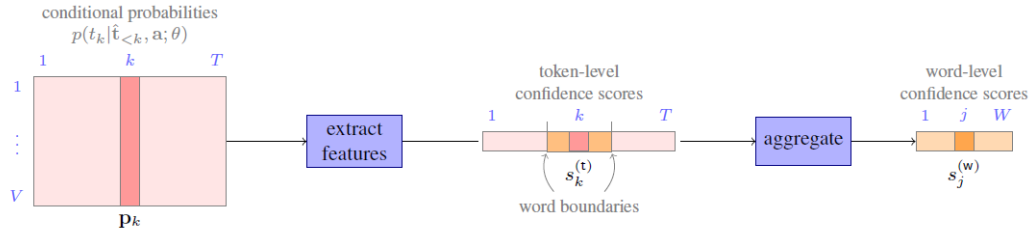


FIGURE 8.2: End-to-end word-level confidence feature extraction pipeline from Wav2vec2 model outputs.

Token-Level Feature Extraction

Two features are computed from the token distributions:

- **Log-Max Probability:** $s^{(t)} = \log \max p$, representing the log-probability of the most likely token.
- **Negative Entropy:** $s^{(t)} = -\sum p \log p$, where lower entropy indicates more confident predictions.

Aggregation to Word-Level

We aggregate the token-level scores into word-level scores $s^{(w)}$ using the following methods:

- **Sum:** Total of token-level scores within the word boundary.
- **Average:** Mean confidence score per token.
- **Minimum:** Lowest confidence score among tokens.

The aggregation accounts for the length of the word and the distribution shape of token confidences.

Temperature Scaling

To calibrate overconfident predictions from E2E models like Wav2vec2, we apply temperature scaling. Given logit activations, we scale them by a temperature parameter

$$p'_k = \text{softmax}\left(\frac{\log(p_k)}{\tau}\right)$$

Here, τ are parameters optimized on a validation set. This approach provides a calibrated estimate of word correctness. We used $\tau = 3$.

8.3 Experiments

We evaluated our confidence features on both English and Hindi test datasets (MPS-English and KV-2000-Hindi), using two metrics: **Miss Rate at a fixed False Positive Rate (FPR) of 5%** and the **Area Under the ROC Curve (AUC)**. The Miss Rate reflects the proportion of actual miscues that were not detected (i.e., false negatives), and a lower value indicates better recall. AUC represents the overall separability of correct vs. miscue classes—higher values are better.

TABLE 8.1: Comparison of confidence-based feature performance using Miss Rate (at 5% FPR) and AUC on the **English MPS test** dataset.

Feature	Miss Rate	AUC
LevDist	40.74	0.89
CostLevDist	39.22	0.89
<i>CostLevDist + E2E Confidence Features</i>		
raw_logmax_min	39.40	0.91
raw_logmax_sum	39.38	0.92
raw_logmax_mean	39.23	0.91
temperature_logmax_min	39.00	0.93
temperature_logmax_sum	37.04	0.92
temperature_logmax_mean	39.19	0.92
raw_entropy_min	39.16	0.88
raw_entropy_sum	38.62	0.92
raw_entropy_mean	39.21	0.91
temperature_entropy_min	39.11	0.91
temperature_entropy_sum	36.81	0.92
temperature_entropy_mean	38.81	0.92

From Table 8.1 and 8.2, we make the following observations:

- **CostLevDist outperforms LevDist** consistently across both languages. This is likely because CostLevDist penalizes confusions between highly similar phonemes (e.g., /s/ vs. /sh/, or vowels) less harshly, making the distance metric more robust to plausible mispronunciations.
- **Temperature scaling improves performance** over raw posteriors. This is because raw Wav2vec2 logits tend to be overconfident; temperature scaling calibrates the confidence distribution, making it more reliable.
- **Entropy-based features outperform log-max features** overall. Entropy captures the uncertainty of predictions more holistically, providing a better estimate of confidence across tokens.

Feature	KV 2000		KV 812	
	Miss Rate	AUC	Miss Rate	AUC
LevDist	52.02	0.84	37.20	0.89
CostLevDist	51.50	0.84	36.93	0.89
<i>CostLevDist + E2E Confidence Features</i>				
raw_logmax_min	51.89	0.85	36.78	0.89
raw_logmax_sum	51.77	0.85	36.86	0.89
raw_logmax_mean	50.20	0.83	35.71	0.89
temperature_logmax_min	52.15	0.86	36.43	0.88
temperature_logmax_sum	50.76	0.86	36.81	0.90
temperature_logmax_mean	50.16	0.84	35.76	0.89
raw_entropy_min	50.67	0.84	36.57	0.89
raw_entropy_sum	51.48	0.85	36.86	0.89
raw_entropy_mean	50.14	0.83	35.33	0.88
temperature_entropy_min	49.09	0.84	35.71	0.90
temperature_entropy_sum	50.14	0.85	36.31	0.90
temperature_entropy_mean	50.06	0.84	35.74	0.89

TABLE 8.2: Comparison of confidence-based feature performance using Miss Rate (at 5% FPR) and AUC across Hindi KV-2000 and KV 812 test datasets.

- **Aggregation strategies differ by language.** For English, **sum** aggregation yields the best performance, likely due to longer word lengths or more stable phoneme sequences. For Hindi, however, no single aggregation clearly dominates, reflecting the greater variability in word structure or pronunciation.
- The best feature for English was **CostLevDist + temperature-scaled entropy sum**, reducing the Miss Rate from 39.22% to **36.81%**. For Hindi, the best configuration was **CostLevDist + temperature-scaled entropy min**, reducing Miss Rate from 51.49% to **49.08%** for KV 2000 and **CostLevDist + raw entropy mean**, reducing Miss Rate from 36.927% to **35.329%** for KV 812.

Overall, these results show that combining edit-distance features with calibrated E2E confidence scores significantly improves our ability to detect reading miscues. These

features help the classifier not only recognize deviations from canonical phoneme sequences but also assess how confidently those predictions were made—enhancing robustness in challenging conditions like children’s speech.

This concludes our exploration of confidence features. In the next chapter, we will evaluate how these features—when combined with hybrid ASR features—impact overall reading miscue detection performance within our full 2-pass system.

Chapter 9

Miscue Detection Performance

As discussed earlier in Chapter 4, all words labeled as “Correct” by the first-pass system were passed to the second-pass system to determine whether they were miscues or correctly pronounced. In the second-pass system, we have two different model : Hybrid ASR and wav2vec2. Each of the model calculate the word level local confidence feature with respect to canonical word. In this section, we describe the experiments conducted to evaluate the effectiveness of combining Hybrid ASR features and Wav2vec2-based features for reading miscue detection. Our primary goal is to leverage the complementary strengths of these two systems to achieve improved performance in terms of miss rate and Area Under Curve (AUC).

In the interest of combining features, we use the logistic regression classifier with balanced bagging for ensembling. The Test datasets were tested in 6-fold Cross validation.

In our experiments, we fixed the acceptable False Positive Rate (FPR) at 5%, a common benchmark for practical miscue detection scenarios. At this fixed FPR, we

aim to minimize the miss rate, thereby ensuring that our system accurately captures as many true miscues as possible.

To evaluate the effectiveness of the combined approach, we use Detection Error Tradeoff (DET) curves, which provide a graphical representation of the trade-off between FPR and Miss Rate. Unlike the Receiver Operating Characteristic (ROC) curve, which plots True Positive Rate against False Positive Rate, DET curves plot miss rate (False Negative Rate) against FPR on logarithmic scales. DET curves are particularly suitable for visualizing system performance when the focus is on minimizing missed detections at a fixed acceptable FPR.

English Language Experiments

Table 9.1 presents the miss rate and AUC for the MPS English test dataset, categorized by feature type and their combinations. Among the Hybrid ASR features, the best-performing GOP-based feature (Min GOP) achieved a miss rate of 39.66% and an AUC of 0.89. Wav2vec2-based features improved upon these results, with the best standalone feature (CostLev + temperature entropy sum) achieving a lower miss rate of 36.84% and an AUC of 0.92.

Wav2vec2 features perform better than Hybrid ASR features. As wav2vec2 has very strong AM influence which give lower PER, hence phone edit distance with confidence feature perform better. However we can see that Min GOP feature is still somewhat comparable to wav2vec2's CostLev feature.

Feature fusion provided further improvements. Specifically, combining the three best-performing features (Fusion-best 3) reduced the miss rate to 32.36%. Using all available features (Fusion-All) yielded the best overall performance, with a miss rate of 31.07% and an AUC of 0.93. This clearly demonstrates that feature fusion

significantly enhances miscue detection capabilities by integrating complementary information from Hybrid ASR and Wav2vec2 feature.

TABLE 9.1: **English MPS Test Dataset :** Miss rate at FPR=5% and AUC for features across different classes and their combinations. Bold font: best in class, (R: raw; T: temperature, L: logmax, E: entropy; S: sum)

Feature	Miss rate(%)	AUC
Lattice-based		
LRT	61.04	0.67
Hypotheses Density	58.37	0.72
Posterior	58.14	0.71
GOP-based		
LLR	49.80	0.87
Mean GOP	47.01	0.88
Min GOP	39.66	0.89
Phone edit distance (wav2vec2)		
Lev	40.74	0.89
CostLev	39.22	0.89
CostLev + raw_logmax_sum	39.39	0.91
CostLev + raw_entropy_sum	38.63	0.92
CostLev + temprature_logmax_sum	37.02	0.92
CostLev + temprature_entropy_sum	36.84	0.92
Feature fusion		
Fusion-best 3	32.36	0.92
Fusion-All	31.07	0.93

Figure 9.1 illustrates the Detection Error Tradeoff (DET) curve comparing the performance of different confidence features and their combinations on the MPS English dataset. The horizontal axis represents the False Positive Rate (FPR), while the vertical axis indicates the False Negative Rate (Misses), both plotted on a logarithmic scale. The leftmost red circle marks (**LM on Prompts**) the performance using only the first-pass system with a Hybrid ASR whose language model (LM) is trained on the canonical story prompts. This setup shows a very high miss rate (72%) due to

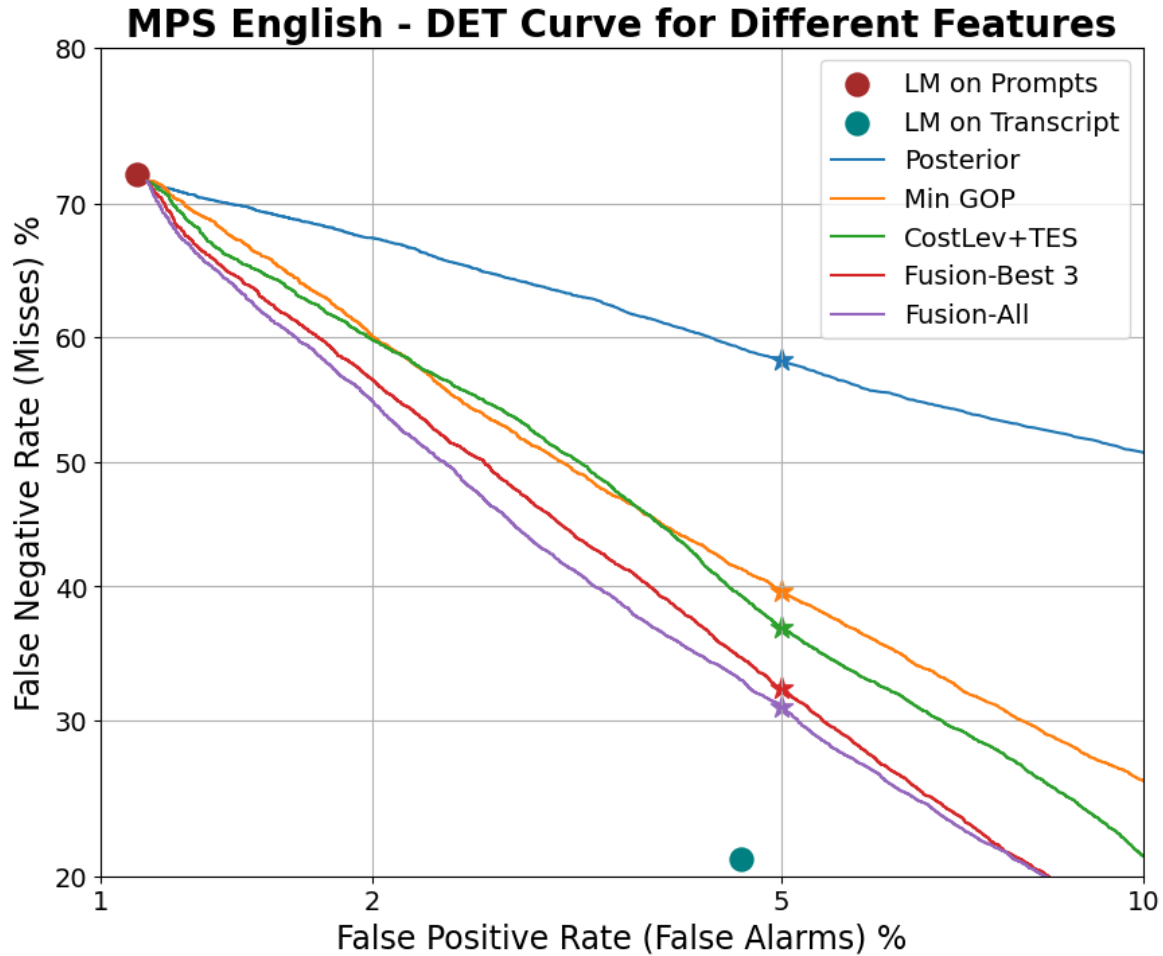


FIGURE 9.1: Detection Error Tradeoff (DET) curve illustrating miscue detection performance on the MPS English dataset. Each curve represents a different confidence feature or feature fusion method evaluated in the second-pass system. The curves are plotted with Miss Rate (False Negative Rate) against False Positive Rate (FPR), with a focus on performance at the operational point of 5% FPR.

strong language model constraints, which tend to mask miscues by favoring expected sequences.

In contrast, the teal circle (**LM on Transcript**) represents the LM trained directly on manual transcriptions of student speech, yielding a significantly lower miss rate. However, such a setup is impractical in deployment since the ground truth is unknown beforehand. The purpose of the second-pass system is to bridge this gap by combining features that generalize better.

Among single-feature curves, the posterior-based score derived from the hybrid word lattice performs poorly, as the lattice is often sparse and therefore not informative in the context of the strongly constrained LM of reading prompts. The Min-GOP feature enhances the trade-off by leveraging localized confidence scores computed relative to the canonical word, leading to more accurate identification of subtle miscues. Notably, Wav2vec2-derived features such as CostLev+TES (Cost based phonetic distance + temperature entropy sum) perform even better due to their stronger acoustic modeling.

Further improvements are observed through feature fusion. The fusion of the three best features (Fusion-Best 3) achieves a considerable reduction in miss rate. The best performance is obtained through Fusion-All, where all features are combined. This fusion significantly lowers the miss rate at 5% FPR, confirming that Hybrid ASR and Wav2vec2 features offer complementary information.

Hindi Language Experiments

Table 9.2 summarizes the performance on two distinct Hindi datasets: KV-2000 and KV-812. Similar trends to the English experiments are observed. Among the Hybrid ASR features, the Min GOP feature again performs best, achieving a miss rate of 52.40% (KV-2000) and 48.33% (KV-812) with AUC values of 0.85 and 0.85 respectively.

Wav2vec2 features consistently outperformed the Hybrid ASR features across both datasets. The best Wav2vec2-based feature (CostLev + temperature entropy min) achieved a miss rate of 49.08% for KV-2000 and (CostLev + raw entropy mean) achieved a miss rate of 35.71% for KV-812, highlighting substantial improvements in detecting miscues.

Feature fusion again showed significant performance gains. For KV-2000, Fusion-All achieved the best miss rate of 42.48% with an AUC of 0.88. Similarly, for KV-812, Fusion-All produced the lowest miss rate of 33.34% and the highest AUC of 0.91. This further confirms the advantage of combining Hybrid ASR and Wav2vec2 features for robust miscue detection in multilingual scenarios.

TABLE 9.2: **Hindi KV-2000 and KV-812 Test Datasets:** Miss rate and AUC at FPR=5% for different features and their combinations on Prompt LM. Bold font: best in class (within each dataset).

Feature	KV-2000		KV-812	
	Miss Rate (%)	AUC	Miss Rate (%)	AUC
Lattice-based				
LRT	60.12	0.72	53.27	0.70
Hypotheses Density	62.00	0.67	52.76	0.70
Posterior	58.29	0.71	53.57	0.70
GOP-based				
LLR	54.11	0.85	52.34	0.72
Mean GOP	56.92	0.83	51.63	0.72
Min GOP	52.41	0.85	48.33	0.86
Phone edit distance (wav2vec2)				
Lev	52.02	0.84	37.20	0.89
CostLev	51.50	0.83	36.93	0.89
CostLev + raw_logmax_mean	50.20	0.83	35.71	0.89
CostLev + raw_entropy_mean	50.14	0.83	35.33	0.88
CostLev + temprature_logmax_mean	50.16	0.84	35.76	0.89
CostLev + temprature_entropy_min	49.09	0.84	35.71	0.90
Feature fusion				
Fusion-Best-3	44.77	0.88	33.90	0.91
Fusion-All	42.49	0.88	33.35	0.91

Figures 9.2 and 9.3 present the Detection Error Tradeoff (DET) curves for the KV-2000 and KV-812 Hindi test datasets, respectively. These plots visualize the trade-off between false positive rate (FPR) and miss rate (false negatives) for various confidence features and their combinations. As in the English experiments, the baseline LM-on-Prompts point starts with a very high miss rate (around 71% for

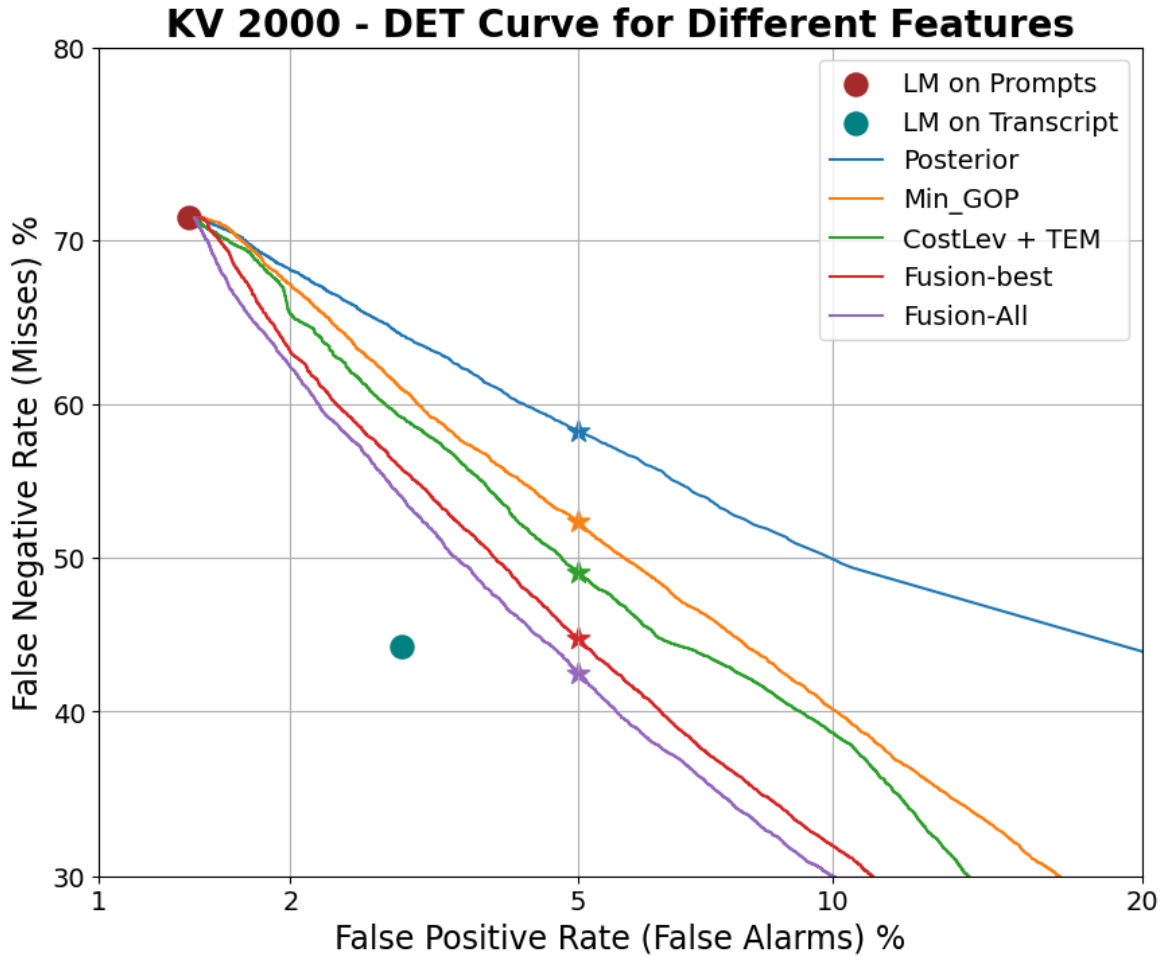


FIGURE 9.2: Detection Error Tradeoff (DET) curve comparing different features for miscue detection on the KV-2000 Hindi test set. Fusion-All achieves the lowest miss rate at 5% FPR, indicating the benefit of combining features from both Hybrid ASR and Wav2vec2-based systems.

KV-2000 and 55% for KV-812), reflecting the limited ability of the first-pass system to detect miscues when constrained by canonical language models.

The LM-on-Transcript point—although not usable in practice—demonstrates the upper-bound performance when the system is trained on exact student transcriptions. Notably, the KV-2000 dataset appears more challenging than KV-812, as reflected by consistently higher miss rates across all features.

For actual feature-based approaches, the posterior scores derived from hybrid ASR

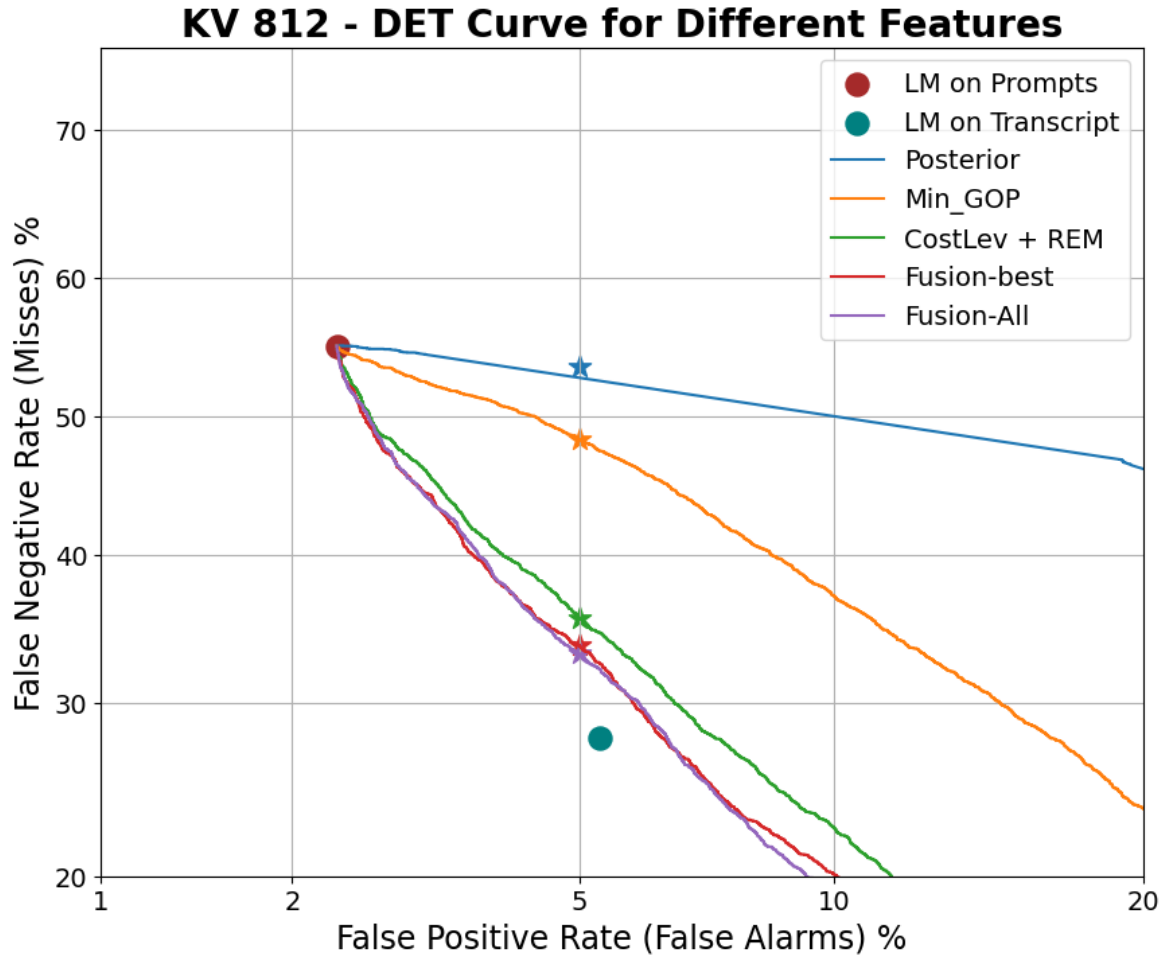


FIGURE 9.3: DET curve showing miscue detection performance for the KV-812 Hindi test set. Similar to KV-2000, Fusion-All outperforms individual features and fusion-best configurations, demonstrating robust generalization across different testset.

lattices yield poor trade-offs, primarily due to sparse alignments under strong LM constraints. Among the hybrid features, Min-GOP consistently provides the best results on both dataset. However, Wav2vec2-based phonetic distance features decisively outperform Min-GOP across both datasets. In particular, variants like CostLev combined with temperature-scaled entropy achieve significantly lower miss rates, highlighting Wav2vec2's superior sensitivity to subtle pronunciation errors and acoustic deviations. This demonstrates the clear advantage of end-to-end acoustic modeling over traditional hybrid approaches for fine-grained miscue detection.

Feature fusion methods further improve detection accuracy. For both datasets as in Table 9.2, Fusion-Best-3 reduces the miss rate considerably, while Fusion-All achieves the lowest miss rate (42.5% on KV-2000 and 33.3% on KV-812) at 5% FPR, confirming the complementary nature of confidence features extracted from both hybrid and Wav2vec2-based systems. The consistent performance across datasets also indicates strong generalization to different stories text.

Chapter 10

Discussion and Analysis

In this chapter, we discuss the performance of the Wav2Vec2 model, highlighting its advantages and limitations. We also provide a breakdown of the two-pass miscue detection performance, with detailed examples, to better understand its effectiveness and areas for improvement.

Wav2vec2 Phone Prediction

To demonstrate the phonetic prediction capabilities of Wav2vec2, we present two examples—one in Hindi and one in English. These examples illustrate how accurately Wav2vec2 can capture phone sequences, even in the absence of an external language model (LM). Here we have used best performing w2v model for each language. Used Hindi w2v model is AI4B SSL IITM + AIS + Pseudolabelled_PRC (B+M+E) → KV Train Hindi (Fine-tuned) as in table 7.6 and English w2v Model is Wav2vec2-XLSR (IITM → WPP (only TL) fine-tuning) as in table 6.1. Hybrid ASR model decode the text in 1st pass, where we used best performing model as in

table 7.6 for Hindi and table 6.1 for English and Hybrid ASR LM model is trained on the canonical text.

Hindi:

Canonical Text : सदाचारी बनने के लिए ईमानदार होना आवश्यक है।

Hybrid ASR Decoded Text : सदाचारी बनने के लिए ईमानदार होना आवश्यक है।

Manual Transcription (MT) Text : सदाचार बनने के लिए ईमानदार होना आवश्यक है।

MT Phone : s aa d AA c AA r * b aa n a n ee * k ee *

l ii ee * II m AA n aa d AA r * hh oo n AA *

AA w aa sh y aa k * hh ee *

W2V Prediction : s aa d AA c AA r * b aa n a n ee * k ee *

l ii ee * II m AA n aa d AA r * hh oo n AA *

AA w aa sh y aa k * hh ee

In this Hindi example, the canonical word given to the student was सदाचारी, but it was actually uttered as सदाचार. The Hybrid ASR decoded the utterance as the canonical word due to its tightly constrained language model, which biases decoding toward expected vocabulary. As a result, this miscue remained undetected in the first pass.

In contrast, Wav2vec2 accurately captured the spoken phones corresponding to सदा-चार, without relying on any external language model. This highlights the strength of its acoustic model (AM), which is capable of distinguishing subtle phonetic variations. Interestingly, despite the absence of an LM, the Wav2vec2 model implicitly

learns language patterns during training, enabling it to generate accurate phone sequences even in such edge cases.

English:

Canonical Text : In the evening, the **shepherd** took them back home. He **counted** the sheep.

Hybrid ASR Decoded Text : in the evening the the **shepherd** took them back home he **counted** the sheep.

Manual Transcription (MT) Text : in the evening the the **शफड** took them back home he **count** the sheep.

MT Phone : IH N * DH AH * IY V N IH NG * SIL * DH AH * SIL * DH AH *
SH AH F AH D * SIL * T UH K * SIL * DH EH M * SIL * B AE K
 * SIL * HH OW M * SIL * HH IY * **K AW N T** * DH AH *
 SH IY P *

W2V Prediction : IH N * DH AH * IY V N IH NG * SIL * DH AH *
SH AH F AH D * T UH K * SIL * DH EH M * B AE K * SIL *
 HH OW M * SIL * HH IY * **K AW N T** * DH AH * SH IY P *

In the English example, the student mispronounced the word **shepherd** as **शफड**, and shortened **counted** to **count**. The Hybrid ASR decoded both words as their canonical forms due to strong linguistic priors—favoring lexicon-constrained decoding and grammatical plausibility.

However, the Wav2vec2 model correctly predicted the actual phones uttered, including the uncommon and mispronounced form of शफड, and also count instead of counted. This reinforces the strength of Wav2vec2’s AM in detecting miscues, particularly when the pronunciation deviates from standard patterns.

One notable observation in this example is that Wav2vec2 omitted the second occurrence of “the”, which the student repeated. This insertion was captured in the decoded text by the Hybrid ASR but not by Wav2vec2.

Confusion Matrix

Figures 10.1 and 10.2 present the phone-level confusion matrices for the Wav2vec2 model evaluated on the MPS English and KV-2000 Hindi datasets, respectively. Each matrix illustrates the frequency with which each ground truth phone (y-axis) is predicted as another phone (x-axis). These confusion patterns were further utilized to construct the substitution cost matrix for the *Cost-Levenshtein Distance* (*Costlevdist*) feature, enabling the system to weight phonetic errors based on actual confusion rates.

For English (Figure 10.1), we observe consistent confusion between several phonetically similar pairs such as:

- /IH/ and /IY/, /EH/ and /AE/ (short vs. long or tense vs. lax vowels),
- /S/ and /Z/ (voiceless vs. voiced alveolar fricatives)
- /V/ and /W/ (voiced fricatives).

These confusions align with common phonetic ambiguities in children’s speech, especially under disfluency or accent variation.

In the Hindi confusion matrix (Figure 10.2), the model similarly struggles with:

- Multiple vowel variants such as /aa/, /ii/, /ee/, /oo/, and nasalized forms,
- Other pairs like /s/ vs. /sh/.

These observations show that while Wav2vec2 performs well acoustically, it is vulnerable to fine-grained phonetic distinctions. This is especially evident in noisy conditions or low-WCPM (Words Correct Per Minute) speech, where pronunciation variability is high. Moreover, this reinforces the need for phoneme-specific substitution costs to be integrated into downstream miscue detection systems.

Both matrices are computed from Wav2vec2 outputs using the English phone set (IITM) and Hindi superset phone set (IITB), as described in Appendix 11.1.

Reading miscue detection analysis

We reduced the miss rate at a fixed false positive rate (FPR) and achieved strong overall performance using Wav2vec2. However, to gain deeper insights into the model’s behavior, we further analyze its performance across different types of miscues and error categories.

Table 10.1 presents the Phone Error Rate (PER) breakdown by prediction type for the two-stage reading miscue detection system, evaluated on the KV Hindi-2000 and MPS English datasets. The overall PERs are 4.45% for Hindi and 9.67% for English, indicating effective general performance. However, a deeper analysis reveals that the highest error rates occur in the False Negative (FN) category, where the system fails to identify actual miscues. Specifically, the FN PER reaches 22.44% for Hindi

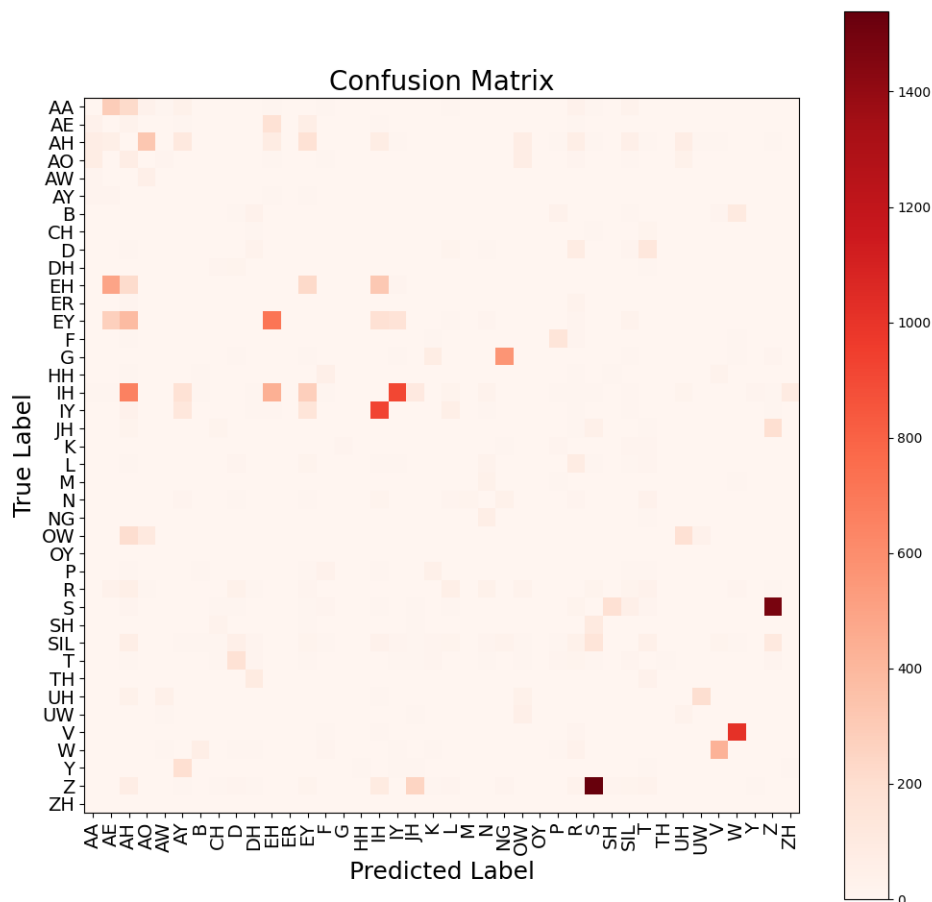


FIGURE 10.1: Confusion matrix of phone predictions on the MPS English dataset.

and 40.74% for English, showing that undetected miscues contribute significantly to overall errors.

True Positives (TP), where the system correctly identifies miscues, also show elevated PERs—18.11% for Hindi and 32.82% for English—highlighting residual phonetic errors even within correctly detected miscue segments. In contrast, True Negatives (TN), where correctly pronounced words are accurately classified as correct, have much lower PERs (3.58% for Hindi and 8.64% for English), suggesting the model

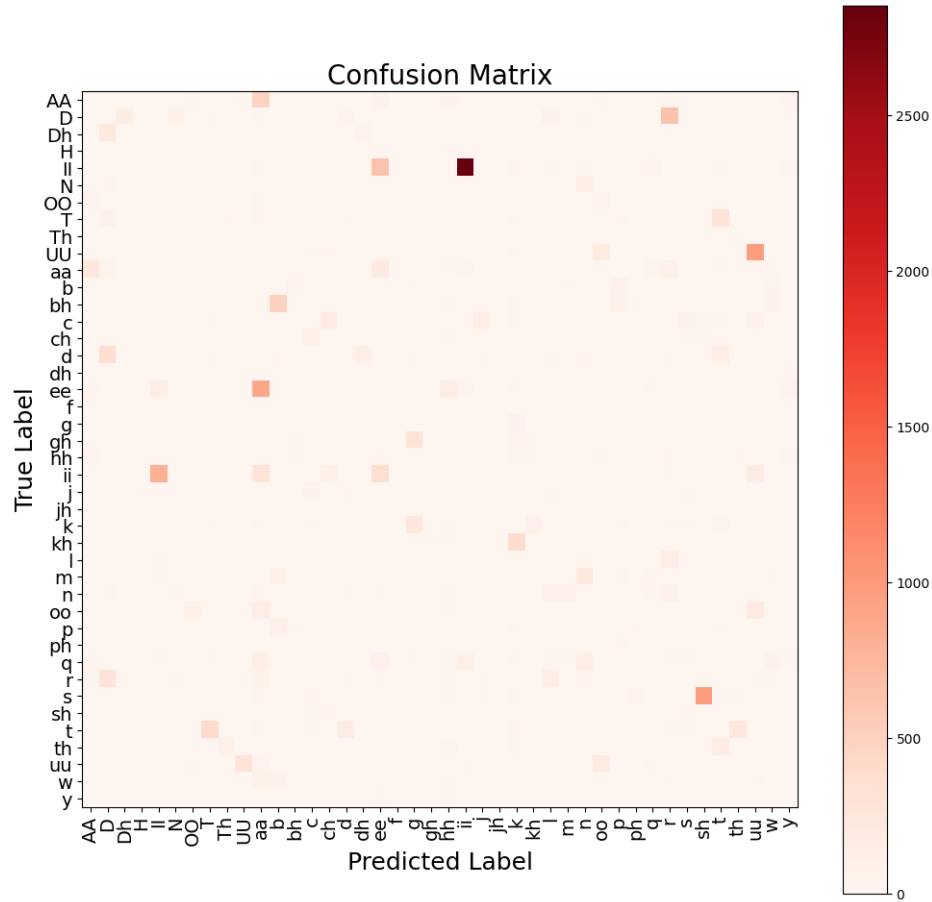


FIGURE 10.2: Confusion matrix of phone predictions on the KV-2000 Hindi dataset.

is more confident and accurate in handling non-miscue segments. False Positives (FP), where correct words are incorrectly flagged as miscues, contribute moderate error rates (4.12% and 9.16%), suggesting a smaller but non-negligible source of error. These findings indicate that while the system is relatively reliable in detecting correctly read segments, there remains significant room for improvement in detecting and transcribing miscues.

TABLE 10.1: PER (%) by Prediction Type after 2-stage Reading miscue System, where only correctly marked words from the first-pass system are forwarded to the second pass. Overall PER on KV-2000 Hindi and MPS English is 4.45 and 9.67 respectively. Hindi model is AI4B SSL IITM + AIS + Pseudolabelled_PRC (B+M+E) $\rightarrow$ KV Train Hindi (Fine-tuned) as in table 7.6 and English Model is Wav2vec2-XLSR (IITM $\rightarrow$ WPP (only TL) fine-tuning) as in table 6.1. Cell values show PER and the proportion of that prediction type in the test set (%)

Condition	Type	KV-2000	MPS English
MT Label = s & Classifier output = c	False Negative (FN): Mislabelled as correct	22.44 (3.78%)	40.74 (3.17%)
MT Label = s & Classifier output = s	True Positive (TP): Correctly predicted miscue	18.11 (5.16%)	32.82 (7.06%)
MT Label = c & Classifier output = c	True Negative (TN): Correctly predicted correct	3.58 (86.51%)	8.64 (85.27%)
MT Label = c & Classifier output = s	False Positive (FP): Incorrectly predicted miscue	4.12 (4.55%)	9.16 (4.48%)

Limitation of wav2vec2

While Wav2vec2 significantly outperforms traditional Hybrid ASR models in terms of Phone Error Rate (PER), it exhibits notable limitations in fine-grained timing alignment, especially at the word and phone levels. Figure 10.3 illustrates a case where Wav2vec2 fails to generate accurate phone-level boundaries for the word "once". Although the phonetic sequence [W, AH, N, S] is correctly predicted, the model aligns all phones to nearly a single frame, resulting in poor temporal localization.

This contrasts sharply with the Hybrid ASR system, which leverages Hidden Markov Models (HMMs) and provides more precise alignment information, as seen in the corresponding phoneme segments where phone boundaries are temporally well distributed. Such precise alignments are crucial in educational ASR tasks like miscue

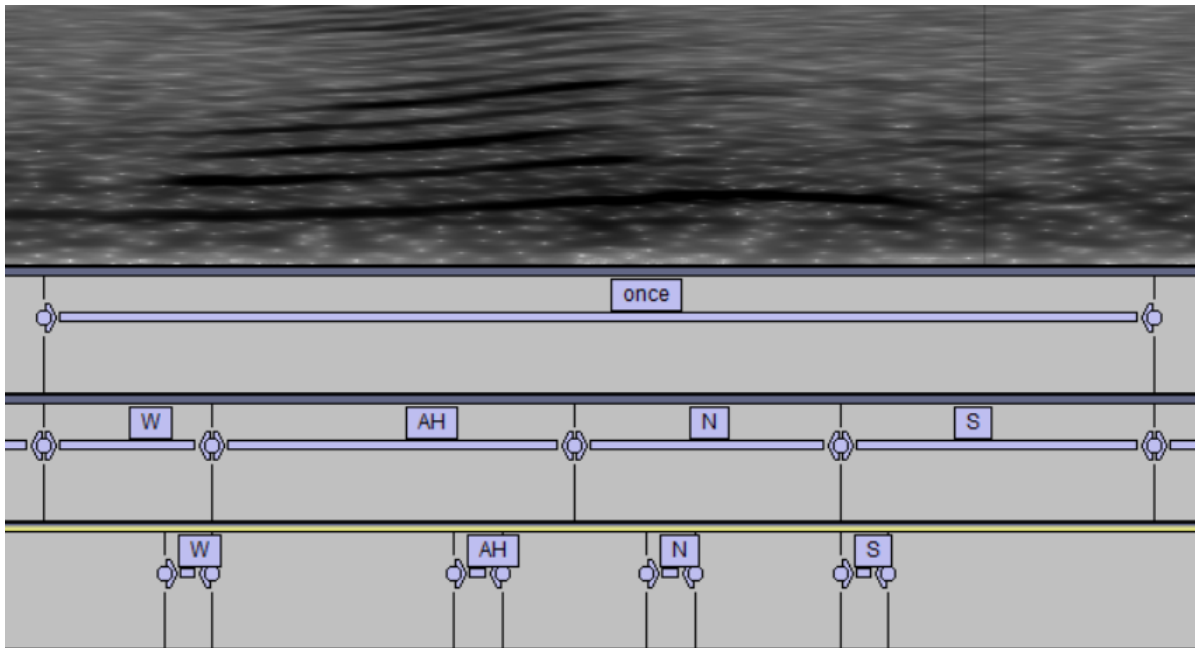


FIGURE 10.3: 1st : Word boundary, 2nd : Phone boundary from Hybrid ASR system and 3rd : Wav2vec2 Prediction boundary

detection, where identifying when a miscue occurs is as important as detecting what the miscue is.

Moreover, Wav2vec2 occasionally omits entire words in low-quality or noisy recordings, particularly for struggling readers with low Words Correct Per Minute (WCPM). This behavior is problematic when the target population includes early-grade children with variable fluency and pronunciation patterns. While Wav2vec2 is highly effective for fluent readers and clean recordings—often catching subtle phone-level errors—it lacks robustness in cases of hesitations, disfluencies, or low signal quality.

Therefore, in our two-stage system, Wav2vec2 features are highly valuable for detecting fine-grained miscues in the second pass, but Hybrid ASR remains beneficial in the first pass for establishing reliable time-aligned word and phone boundaries.

Chapter 11

Conclusion and Future Work

11.1 Conclusion

This thesis addressed the critical challenge of improving miscue detection in Oral Reading Fluency (ORF) assessments using state-of-the-art ASR techniques, specifically leveraging the Wav2vec2 model. Through comprehensive experiments, we demonstrated substantial improvements in phone recognition and reading miscue detection accuracy across Hindi and English datasets.

Our work initially involved constructing robust phone recognition models based on the Wav2vec2 architecture. Training strategies included fine-tuning self-supervised pretrained models using labeled adult speech data and further adapting these models to children’s speech through labeled and pseudo-labeled datasets.

When trained solely on adult datasets, the Wav2vec2 model achieved a Phone Error Rate (PER) of 23.24% on the MPS English test set. Adding labeled children’s data reduced the PER significantly to 9.67%, demonstrating the importance of using in-domain children’s speech for model adaptation. For Hindi, the model trained only

on adult data achieved PERs of 10.12% on KV-2000 and 9.80% on KV-812. By incorporating both pseudo-labeled and labeled children’s speech data, we further reduced the PER to 4.45% for KV-2000 and 6.62% for KV-812, highlighting the effectiveness of our adaptation strategy and the superior acoustic modeling capabilities of Wav2vec2. Overall, Wav2vec2 consistently outperformed the hybrid ASR system in phone recognition across all configurations.

Moreover, we utilized pseudo-labeling techniques combined with careful filtering based on Connectionist Temporal Classification (CTC) loss and Probability Ratio-based Confidence (PRC) scores. This approach allowed the effective utilization of a substantial amount of unlabeled children’s speech data, contributing further to model generalization and adaptation. Ablation studies revealed PRC-based filtering to be particularly effective, highlighting the importance of reliable confidence estimation for training data selection.

Additionally, we investigated the impact of incorporating labeled Hindi children’s data on both overlapping and non-overlapping evaluation sets with respect to story text. On the overlapping test set (KV-2000), which shares stories with the training data, the PER dropped to 4.45% through further fine-tuning, indicating effective domain adaptation. In contrast, for the non-overlapping test set (KV-812), where story prompts differ from those in training, the best PER of 6.62% was achieved using combined training of all data, rather than fine-tuning alone. This highlights the importance of broader data diversity and generalization for robust performance on unseen prompts.

Subsequently, we proposed a two-pass miscue detection framework that integrates hybrid ASR for precise word-level alignment and Wav2vec2 for phone-level predictions and confidence estimation. This approach leverages the complementary

strengths of both architectures: the robust alignment capabilities of hybrid models and the superior acoustic modeling of Wav2vec2. Confidence features derived from Wav2vec2 outputs—such as Levenshtein distance (LevDist, CostLevDist) and end-to-end confidence metrics—proved highly effective in identifying miscues. Our experiments demonstrated that Wav2vec2-based features consistently outperformed those from the hybrid ASR system. Moreover, fusing features from both systems led to further reductions in miss rate, highlighting the complementary nature of the information captured by each model.

In summary, this thesis uses Wav2vec2 for miscue detection in multilingual educational contexts. By combining multilingual phone modeling, pseudo-labeling strategies, and confidence-based feature extraction, our proposed method delivered robust miscue detection performance. This research has laid the groundwork for developing more reliable automated reading assessments, providing meaningful, actionable feedback crucial for pedagogical interventions.

11.2 Future Work

To further improve the accuracy and generalizability of reading miscue detection systems, the following directions are proposed:

1. **Exploring Conformer-based architectures:** Conformer models, which combine convolution and self-attention, are known to outperform standard transformers in speech tasks. Replacing the transformer backbone in Wav2vec2 with conformer encoders may yield superior phoneme discrimination, especially for subtle mispronunciations. [Gulati et al. \(2020\)](#)

2. **Enhancing time-alignment with auxiliary losses:** Although Wav2vec2 with CTC enables sequence prediction, it often lacks fine-grained frame-level alignment. Introducing auxiliary alignment losses—such as monotonic attention or duration prediction losses—could improve time resolution, enhancing miscue localization in spontaneous reading [TEYTAUT et al. \(2022\)](#); [Goswami et al. \(2025\)](#); [Long et al. \(2019\)](#).
3. **Robust training for disfluent or noisy speech:** Children with reading difficulties or recordings from classrooms may contain hesitations, self-corrections, or background noise. Training with synthetic noise augmentation, disfluency injection can enhance model robustness to such real-world variabilities.
4. **Improving recognition of OOV and complex words:** Miscues are more frequent in complex or less frequent words. Specialized sub-networks can help Wav2vec2 better learn OOV representations. Additionally, an OOV detector could guide focused miscue detection on critical segments.
5. **Multilingual generalization:** Leveraging multilingual training and expanding labeled and pseudo-labeled datasets to include more Indian languages could enable scalable assessment systems with minimal adaptation cost.
6. **Parameter-efficient fine-tuning:** Techniques such as Low-Rank Adaptation (LoRA) [Hu et al. \(2021\)](#), adapters and prompt-tuning [Pratap et al. \(2024\)](#) offer efficient alternatives for domain adaptation to story-specific text. These methods can substantially reduce the computational cost of fine-tuning, particularly when adapting models across multiple languages or diverse age groups.
7. **Confidence-based data selection and curriculum learning:** Future studies could explore confidence-driven curriculum learning, where training data is

introduced progressively based on CTC loss or entropy measures, enhancing stability and generalization.

8. **Fine-grained diagnostic feedback and learner-specific intervention:**

Beyond binary miscue detection, future work could focus on generating detailed diagnostic outputs by classifying the type of miscue and identifying specific phonetic deviations from the canonical form. Learners could then be grouped based on their predominant error patterns—such as common confusion between particular phonemes or frequent substitutions. This would enable personalized feedback and targeted intervention strategies, such as recommending phonetically similar or contrastive word lists for practice. Such an approach would enhance the pedagogical value of the system by not only detecting errors but also guiding students toward meaningful improvement.

9. **Evaluating HuBERT for English models:** For English-only modeling, exploring HuBERT [Hsu et al. \(2021\)](#) as an alternative to XLSR may yield improved performance. Since XLSR is trained on multilingual corpora, it may introduce language interference. HuBERT, trained solely on English data, could provide more language-specific acoustic representations, improving miscue sensitivity and phoneme recognition in English reading tasks.

10. **Subword-based tokenization strategies:** Currently, the w2v uses monophone-based tokenization. In future work, bi-phone or tri-phone units could be explored as basic tokens. This could reduce reliance on explicit word-boundary symbols (e.g., *) by capturing richer subword transitions. Such modeling—similar to Whisper’s tokenization strategy—could enable smoother decoding and better capture of coarticulation patterns common in children’s speech.

Appendix of Phone Mappings

IITB	IITM English	IITM Hindi	Word Usage
aa	AH	a	अगली
AA	AA	aa	आगे
ae	AE	ei	एहसास
aw	AW	aa uu	औद्योगीकरण
ay	AY	aa y	ऐतिहासिक
b	B	b	बादल
bh	B HH	bh	कभी
c	CH	c	चौका
ch	CH HH	ch	कुछ
d	DH	d	खरीद
dh	DH HH	dh	गंगाधर
D	D	dx	गाड़ी
Dh	D HH	dxh	ढल
ee	EY	ee	आए
ex	EH	ee	इंस्पेक्टर
f	F	f	फ़ोन
g	G	g	गमला
gh	G HH	gh	घर
hh	HH	h	हाथी
ii	IH	i	चिराग
II	IY	ii	खरीद
j	JH	j	जल्दी
jh	JH HH	jh	मुझको
jz	ZH	j	सीझाद
k	K	k	कुछ
kh	K HH	kh	खरीद
l	L	l	लाल
m	M	m	मुजिक
n	N	n	नदी
ng	NG	n g	Running
N	N N	nx	विषणु
oo	OW	o	मुझको
OO	AO	ou	और
oy	OY	ou y	voice
p	P	p	up
ph	P HH	ph	फुगे
r	R	r	रंग
s	S	s	सीखे
sh	SH	sh	शरण
t	TH	t	तुम
th	TH	th	थकलेले
T	T	tx	नाटक
Th	T HH	txh	पाठ
uu	UH	u	तुम
UU	UW	uu	बूझ
v	V	w	विषणु
w	W	w	पड़ाव
y	Y	y	येत
z	Z	z	zoo
zh	Z HH	jh	झाडे
H	HH	h	अतः

The ॐ and ॐ and 1 symbols in devnagri are mapped to q to denote nasalisation

TABLE 11.1: Phone mapping from the IITB superset to the IITM English and IITM Hindi phone sets.

Publications and Resources

Publications

- Gothi, R., Kumar, R., Pereira, M., Nayak, N., Rao, P., 2024. A dataset and two pass system for reading miscue detection, in: Interspeech 2024, pp. 4014–4018. doi:10.21437/Interspeech.2024-1639.

Code and Dataset Links

- **GitHub Repositories:**
 - Wav2vec2 Phone Recognition & Reading Miscue Detection:
<https://github.com/RajGothi/Wav2vec2-PhoneRecognition>
 - Levenshtein Alignment using POWER Algorithm:
<https://github.com/nageshnayak/levenshtein-alignment>
- **English Test Dataset (MPS) & Notion Page:**
 - GitHub: https://github.com/DAP-Lab/mps_dataset
 - Notion: <https://tinyurl.com/35x2wvy7>
- **Hindi Test Datasets (KV-2000, KV-812):**
 - Notion: <https://tinyurl.com/RRIS25>

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